Computers in Biology and Medicine 196 (2025) 110922



Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed





Trust and explainability in robotic hand control via adversarial multiple machine learning models with EEG sensor data fusion: A fuzzy decision-making solution☆

A.S. Albahri [a,b], Rula A. Hamid [c], M.E. Alqaysi [d], Z.T. Al-Qaysi [e], O.S. Albahri [f,g], A.H. Alamoodi [h,i,*], Raad Z. Homod [j], Muhammet Deveci [k,l,m], Iman Mohamad Sharaf [n]

[a] Technical Engineering College, Imam Ja'afar Al-Sadiq University (IJSU), Baghdad, Iraq
[b] University of Information Technology and Communications (UOITC), Baghdad, Iraq
[c] College of Business Informatics, University of Information Technology and Communications (UOITC), Baghdad, Iraq
[d] Computer Sciences Department, Computer Science and Information Technology College, Anbar University, Anbar, Iraq
[e] Department of Computer Science, Computer Science and Mathematics College, Tikrit University, Tikrit, Iraq
[f] Computer Techniques Engineering Department, Mazaya University College, Nasiriyah, Iraq
[g] Institute of Innovation, Science and Sustainability (IISS), Federation University, Melbourne Campus, Australia
[h] Applied Science Research Center, Applied Science Private University, Amman, Jordan
[i] GUST Engineering and Applied Innovation Research Center (GEAR), Gulf University for Science and Technology, Mishref, Kuwait
[j] Department of Oil and Gas Engineering, Basra University for Oil and Gas, Iraq
[k] Department of Industrial Engineering, Turkish Naval Academy, National Defence University, 34942 Tuzla, Istanbul, Türkiye
[l] Department of Information Technologies, Western Caspian University, Baku 1001, Azerbaijan
[m] Department of Computer Science and Engineering, College of Informatics, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea
[n] Department of Basic Sciences, Higher Technological Institute, 10th of Ramadan City, Egypt

A R T I C L E I N F O

Keywords:
Feature fusion
Adversarial
Explainability
Evaluation and benchmarking
Machine learning
Fuzzy multicriteria decision-making

A B S T R A C T

In the field of brain–computer interfaces (BCIs), developing a reliable machine learning (ML) model for real-time robotic hand control systems based on motor imagery (MI) brain signals requires substantial research. For this purpose, a set of ML models has been developed and tested to identify robust models via MI sensor data fusion under both nonadversarial and adversarial attack conditions. This paper addresses numerous essential areas, including the development of ML models for electroencephalography (EEG) MI signal datasets, with a focus on proper preprocessing and evaluation under both nonadversarial and adversarial attack conditions. Three phases make up the process. In the first phase, raw MI-EEG datasets from the Graz University BCI competition are identified and preprocessed. The preprocessing encompasses six key stages: EEG-MI signal filtering, segmentation, time–frequency domain feature extraction, merging and labeling, normalization (resulting in Dataset I), and feature fusion (resulting in Dataset II). In the second phase, both datasets are used to develop nine different ML methods and are evaluated via nine performance metrics. These models are trained and tested against adversarial and nonadversarial scenarios. In the third phase, the fuzzy decision by opinion score method (FDOSM) and the multiperspective decision matrix (MPDM) are combined to benchmark the ML models via the fuzzy multicriteria decision-making (MCDM) approach. The random forest (RF) model achieved the best overall performance, with the lowest FDOSM scores: 0.18241 for Dataset I and 0.21636 for Dataset II. A lower FDOSM score means better results across all evaluation criteria. To further assess the developed methodology, the RF model was tested on Dataset III, comprising EEG data from four participants collected via the EMOTIV EPOC. The mean classification accuracy achieved by the RF model was 83 % with standard preprocessing, and it improved to 86 % with the application of feature fusion techniques. Additionally, this study employed the local

☆ This research followed ethical guidelines for research involving human subjects, the first dataset used in this study was publicly accessible and did not require informed consent. For the second dataset, all participants provided informed consent prior to their participation. Data collected was anonymized to ensure confidentiality.

* Corresponding author. Applied Science Research Center, Applied Science Private University, Amman, Jordan.
E-mail addresses: alamoodi.abdullah91@gmail.com (A.H. Alamoodi), muhammetdeveci@gmail.com (M. Deveci).

https://doi.org/10.1016/j.compbiomed.2025.110922
Received 24 May 2024; Received in revised form 5 August 2025; Accepted 7 August 2025
Available online 11 August 2025
0010-4825/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

interpretability model-agnostic explanation (LIME) method to provide an understanding of the RF model's behavior and enhance the interpretability of the results in the context of individual predictions.

1. Introduction

The use of electroencephalography (EEG) sensor data for controlling robotic hands is an exciting and active area of research. This emerging technology aims to create interfaces that enable individuals to control a robotic hand or artificial limb via brainwaves [1]. Using EEG sensor data for robotic hand control involves the use of specialized EEG sensors to record and interpret electrical brain activity. These sensors detect and decode electrical impulses produced by the brain's intent for how the user should execute specified movements (e.g., grasping and manipulating) of the robotic hand [2]. EEG data are processed via machine learning (ML), and the resulting data can be transformed into actions that enable the use of a robotic hand or prosthesis to perform physically complex hand motions.

The significance of using EEG sensor data to control robotic hands has started to emerge in the context of ML [3]. ML technology is an important aspect that enables researchers to train models on EEG data and convert them into commands for robotic hands [4]. Despite their utility, ML technologies are also susceptible to adversarial attacks [5]. Adversarial attacks are an area of research that focuses on attacks against ML algorithms and ways to defend against these attacks [6]. Control of robotic hands to perform tasks on the basis of EEG sensor data is an area where adversarial ML poses a potential threat. The adversary may need to perturb the EEG signal or the ML model so that it has a weakened ability to produce correct control instructions for the robotic hand [7]. Many aspects of robotic hand control development should account for adversarial attacks [8]. It is prudent at this juncture to comment on how many forms of adversarial attacks weaken the robustness of ML models. There are many configurations of adversarial attacks. The majority of configurations in the literature divide the attack into two forms on the basis of how the attacks can be executed: white-box and black-box attacks. Both white- and black-box attacks compromise the robustness of the model, leading to incorrect predictions and undesirable accuracy interpretations. Typically, both black- and white-box attacks are executed to weaken the model's accuracy. In white-box attacks, the attacker has access to the model structure, allowing them to modify the input parameters by adjusting the added perturbations, which can significantly reduce the model's accuracy. The fast gradient sign method (FGSM) and projected gradient descent (PGD) examples of attacks make the models susceptible to hacking attacks if they are not adequately trained on these types of data [9]. The second type of adversarial attack is the black-box attack, where the attacker lacks access to the model's internal details and employs query-based iterative approaches to attack [10]. The black-box attack then weakens the model's robustness, leading to misclassification and further weakening its robustness. Hence, mitigation strategies for both types of adversarial attacks should be employed to train and mitigate their impact, such as adversarial training and robust optimization strategies, which can be utilized to improve the model's robustness [11]. Therefore, by focusing on validating the integrity of EEG sensor data and increasing the robustness of ML models, it will be able to reduce security risks from adversarial attacks and improve robotic hand control [12]. During this time, there is a significant need to study the vulnerability of EEG sensor data and the potential for adversarial attacks to spoof control signals on EEG-based robotic hand control [13]. Understanding these threats and building robust methods for detection are critical for ensuring reliability and security in robotic hand control systems [13].

There are multiple considerations regarding the verification of EEG sensor data integrity and strengthening the robustness of ML models as safeguards against adversarial attacks and for efficient operation of robotic hand control [14,15]. Preprocessing stages serve as the foundation for ML model development, encompassing several pivotal approaches. Given the profound impact of preprocessing on ML models, this stage warrants substantial focus to enhance the real-time operation of robotic hand control systems. Despite the achievement of high-value results after ML model development, questions persist, particularly concerning the opacity of ML models and their intricate interpretability [16]. There are options to explore various preprocessing methods to gain an overarching view of ML models and address the central question in our first hypothesis: "Do distinct preprocessing approaches influence the development of the optimal ML model?" As previously noted, using t-distributed stochastic neighbor e-embedding (t-SNE) appears to be a promising approach for this purpose, providing a more comprehensive view of ML model performance [17]. The final, larger case is the need to determine and evaluate multiple ML models in more complex situations for model exploration and to track real-time model robustness during both normal (nonadversarial) and adversarial attack conditions [15]. Consequently, the development of multiple ML models is imperative, considering various preprocessing approaches. Consequently, following the resolution of the initial hypothesis, multiple ML models emerge, each exhibiting distinct performance evaluation metric values. These models encompass a wide spectrum of ML algorithms, including k-nearest neighbors (kNN), decision trees (DTs), stochastic gradient descent (SGD), support vector machines (SVMs), random forest (RF), gradient boosting (GB), neural networks (NNs), AdaBoost, and logistic regression (LR). Furthermore, the chosen models are more desirable because they are great building blocks for learning purposes, as they are easy to implement and computationally less expensive [18–21]. They are meticulously trained and tested, encompassing diverse preprocessing approaches and spanning both nonadversarial and adversarial attack scenarios. This challenge aligns with the domain of ML theory, where innovative solutions are poised to provide valuable insights and solutions to these complex questions. In this context, this study discusses the second set of hypotheses related to the question of "How can you evaluate and benchmark these models?" The evaluation step involves a thorough assessment against a range of criteria or metrics, allowing comparisons of models to be made and evaluating their strengths and weaknesses. The essential aspect of this process involves establishing weights for each metric, indicating their relative importance. This comparison considers weighing option selections on the basis of established criteria, including metrics related to classification accuracy (Acc), precision, recall, etc. [22]. Within the process, it will also be necessary to conduct trade-off analyses, as the various types of models may outperform one another in one criterion while being lower in others [23]. Therefore, the decision-making process involves conducting a trade-off analysis to determine the best option or combination of options that results in a compromise among these competing criteria [24]. This is not a simple task, as part of the overall process requires considering the relative importance, or weights, associated with each criterion to make an informed decision. In addition, the decision-making process is further complicated by having both benefit and cost criteria. The benefit criterion means that the higher the value is, the better, whereas the cost criterion means that the lower the value is, the better [25]. The implications of this opposition lead to uncertainty in determining the best model, as trade-offs often need to be made between criteria. All the issues related to the second set of hypotheses create a complicated multicriteria decision problem; therefore, multicriteria decision-making (MCDM) has been identified as a useful and necessary tool for this purpose. The MCDM method enables the making of informed decisions for evaluating and selecting an appropriate ML model [26,58,59]. The fuzzy decision by opinion score method (FDOSM) can be a valuable part of MCDM methods when addressing very complex problems [27,28].

FDOSM was developed for assessing metric issues and determining which ML solution is the most highly ranked [29]. Specifically, it considers the idea of having an ideal or best solution, taking into account the level of preference and degree of inconsistency, reduces the number of comparisons needed, employs equal and implicit comparisons, and performs fewer complicated calculations [30]. FDOSM generally overcomes the issues of normalization and weighting in MCDM methods [31]. Most importantly, FDOSM can manage ambiguous and fuzzy data, which are generally identifiable. FDOSM enables the selection of the best ML model, as it is a comprehensive, outcome-based decision that considers several factors and the overall performance of the subjects.

MCDM is a comprehensive approach to decision-making that incorporates multiple objectives and criteria [32]. In terms of attaining the best solutions for Motor Imagery (MI) goals, the MCDM and FDOSM methods offer a systematic means to analyze and assess the best model in terms of several metrics [30]. This capability not only addresses assessment metric challenges but also helps identify the ideal model for the given task, aligning with the objectives of this study.

Additionally, this study aims to explore the trustworthiness of the selected ML model for robotic hand control systems, which is a vital consideration in the realm of artificial intelligence (AI). Trustworthiness in AI applications refers to a multifaceted concept encompassing several key components [33], serving as the foundational cornerstone for the success of AI systems. These trustworthiness components, which are well documented in various fields of research, encompass a range of factors, including reliability, transparency, fairness, privacy, and security [33, 34]. These elements are instrumental in ensuring the reliability and credibility of AI systems. One crucial aspect involves considering adversarial elements within the trustworthiness framework, denoted as the "*Privacy and Data Governance*" component, in which security plays a critical role [35]. The adaptivity of this component aims to safeguard the developed ML model within the AI system from disclosing its capabilities to potential attackers. Technical robustness and safety represent another significant trustworthy component, encompassing protection against security threats; the need for backup plans in case of system failure; and the pursuit of accuracy, reproducibility, and reliability. Therefore, constructing an ML model that considers the ability to defend against adversarial attacks is pivotal for ensuring the trustworthiness of hand control systems. Furthermore, enhancing trustworthiness involves making ML models more explainable, as exemplified by methods such as local interpretability model-agnetics explanations (LELEs) [36]. This explainability plays a pivotal role in bolstering the efficiency and robustness of robotic hand control systems [37].

Finally, according to what has been addressed above, the primary objective of this study is to introduce a novel methodology for the evaluation and benchmarking of multiple ML models. This methodology considers two preprocessing approaches: standard and feature fusion, as well as scenarios involving both nonadversarial and adversarial attack examples. The study places particular emphasis on assessing the outcomes of the proposed methodology, with a focus on validation and explainability for the best-selected ML model, determined through the fuzzy decision-making solution. Within this framework, the three predefined hypotheses can be systematically addressed and discussed, shedding light on critical aspects of ML model development and evaluation. The principal contributions of this study can be summarized as follows:

1. Generation of two meticulously preprocessed MI-EEG datasets through a six-stage processing pipeline, leveraging both standard and feature fusion preprocessing approaches.
2. Development of multiple ML models using preprocessed datasets, taking into account scenarios involving nonadversarial and adversarial attacks for robotic hand control systems.
3. Introduction of a novel fuzzy decision-making framework for evaluating and benchmarking the developed ML models. This framework integrates MPDM and FDOSM to facilitate the selection of the most suitable model.
4. Comprehensive evaluation and validation of the selected ML model using previously unseen MI-EEG data (Dataset III).
5. Integration of the LIME model enhances the trustworthiness of the selected ML model, particularly in adversarial ML scenarios, by providing transparent explanations for model decisions.

These contributions collectively advance the understanding and practical application of ML models in the context of robotic hand control systems, emphasizing both performance and trustworthiness considerations.

2. Methodology

This developed methodology consists of three phases. Phase one (Fig. 1) focuses on the identification and preprocessing of the EEG-ML dataset, resulting in two datasets: Dataset I, which is based on standard preprocessing, and Dataset II, which consists of feature fusion preprocessing.

Furthermore, Fig. 2 provides a visualization of the second and third phases of the methodology. In phase two, multiple ML models for robotic hand control systems are developed using both preprocessed datasets. Two scenarios are addressed when creating the models, namely, nonadversarial attacks and adversarial attacks. In the third phase, the decision-making process is addressed by evaluating and benchmarking the resulting ML models to determine which one is the best. In this phase, the optimal model is identified via FDOSM and MPDM, which review both nonadversarial and adversarial attacks. This comprehensive methodology offers a thorough and structured approach to navigating the challenges of developing and selecting ML models for robotic hand control, ensuring that both trust and robustness are evaluated in the decision-making process.

2.1. Phase 1: Dataset Description and Preprocessing

This stage involves two principal areas: developing multiple reliable and interpretable adversarial ML models and utilizing both standard and fusion preprocessing techniques for robotic hand control via BCIs, which are collected from EEG sensor data. The first stage involves an extensive and detailed discussion of the datasets, encompassing all relevant aspects of the data, including their features, lack of channels, dataset size, and other pertinent details. The next stage involved preprocessing the dataset, a critical phase in the ML pipeline. In this phase, several operations and methods were applied to the dataset to make it useable with ML algorithms while also maximizing its quality and relevance for the task at hand.

2.1.1. BCI EEG-MI dataset

This dataset is part of the tradition of BCI competition datasets and is affiliated with Graz University [38]. Three channels, namely, C3, Cz, and C4, were used to acquire EEG signals for two MI hand movement tasks (right and left) collected from nine subjects at a sampling frequency of 250 Hz. EEG data from 160 trials were collected from the nine subjects while they watched a flat screen and sat in an armchair. Two types of recording sessions were conducted, namely, training without feedback and evaluation with smiley feedback. During the training session, the subjects were given a short warning tone to perform the required MI movement for 4 s, which was based on a pointing arrow presented on a blank screen. However, during the evaluation recording session, the subjects were instructed to move the gray smiley feedback centered on the monitor in either the right or left direction after they were given a short warning beep. The smiley feedback was presented in 4 s, and its color changed to red when the participant moved in the wrong direction and green when they moved in the right direction. Fig. 3 illustrates the timing scheme of the recording technique with and

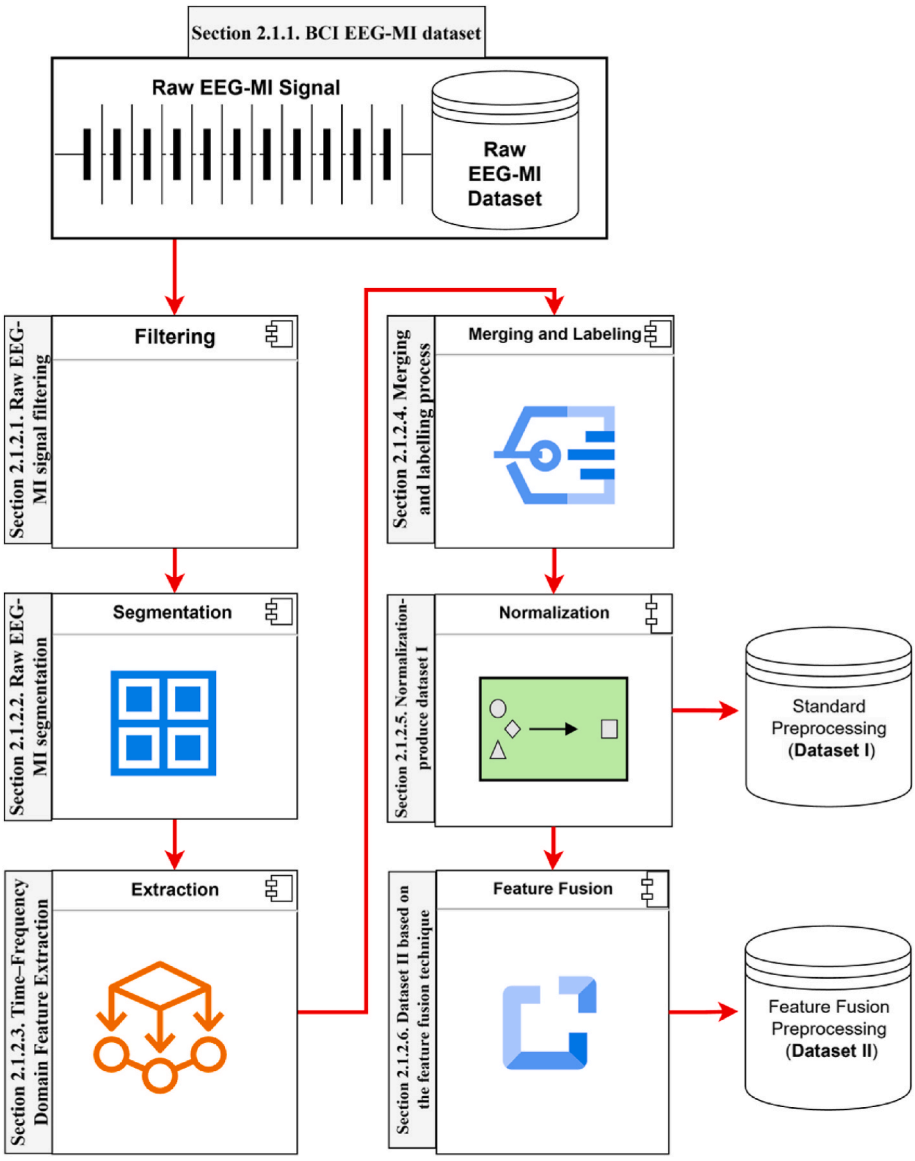


Fig. 1. Phase 1 dataset description and preprocessing of EEG data.

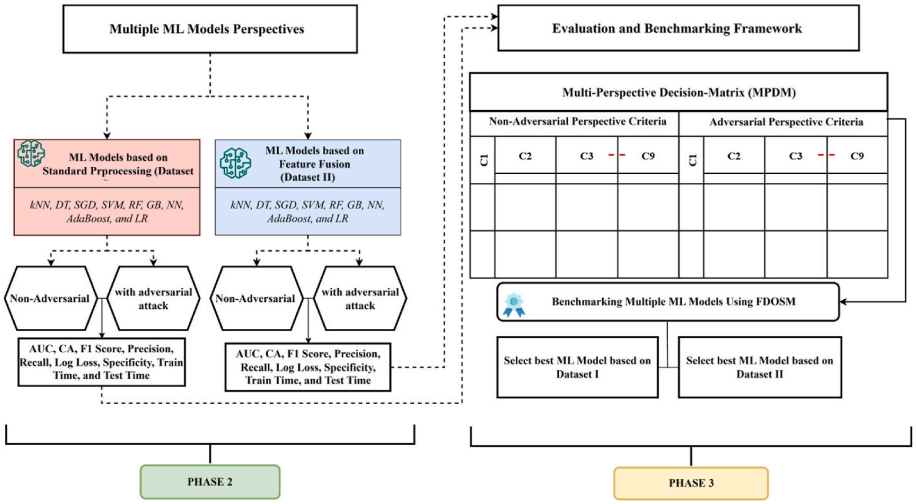
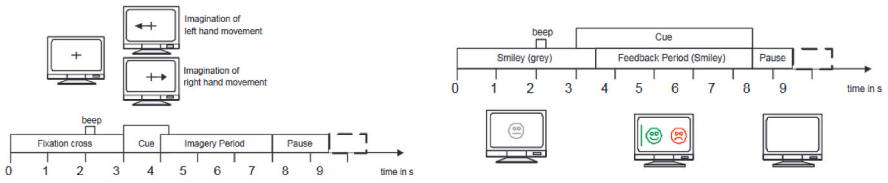


Fig. 2. Phase 2 focuses on developing multiple ML models, and Phase 3 involves evaluating and benchmarking these models.



(a) protocol without feedback (b) protocol with smiley feedback

Fig. 3. Timing scheme of the EEG-MI experiment (a) without feedback, (b) with Smiley feedback [38].

without smiley feedback.

2.1.2. Preprocessing

One approach to managing high-dimensional datasets involves feature selection, which helps remove irrelevant features, simplifying the decoding process. In some cases, features may need to be labeled, and relevant features can be fused to increase model performance by integrating data from multiple sources. Additionally, different data types require various preprocessing techniques. Techniques such as noise removal, scaling, and normalization are essential for datasets containing images.

In the context of developing multiple ML models for robotic hand control, preprocessing of the raw EEG signal is a fundamental and multistage process. This study encompasses six key preprocessing stages: EEG-MI signal filtering, EEG-MI segmentation, time–frequency domain feature extraction, merging and labeling, normalization (resulting in Dataset I), and feature fusion (resulting in Dataset II). The primary objective of this section is to generate two distinct datasets: Dataset I, derived from the standard process, and Dataset II, derived from the feature fusion process. These datasets serve as the foundation for subsequent model development and evaluation. The steps of EEG-MI dataset preprocessing are shown in Algorithm 1.

Algorithm 1: EEG-MI Raw Data Preprocessing

Input:
- EEG-MI Signal
- Time Segment Groups
- Time Frame

Output: *Dataset I and Dataset II*

***Step 1*: EEG-MI Signal Filtering**
i. Define a fourth-order Butterworth filter.
ii. Specify the frequency range of interest (8 Hz–30 Hz).
iii. Apply the Butterworth filter to the EEG-MI signal to remove noise and isolate the desired rhythms.

***Step 2*: EEG-MI Segmentation**
For each specified time frame in Time Segment Groups:
i. Divide the filtered EEG-MI data into nonoverlapping time segments of the specified duration.
ii. Append these time frames to the segmented data.

***Step 3*: Time-Frequency Domain Feature Extraction (using Discrete Wavelet Transform - DWT)**
Utilize the Discrete Wavelet Transform (DWT) to extract relevant coefficients as features from each time frame.

***Step 4*: Merging and Labeling**
i. Merge the DWT features from all time frames into a unified dataset.
ii. Add the movement duration as an additional feature for each data point.
iii. Label the dataset with "movement" as the target class.

***Step 5*: Normalization**
Apply min–max normalization to the features of the labeled dataset to ensure consistent scaling and produce Dataset I.

***Step 6*: Feature Fusion (resulting in Dataset II using t-SNE)**
i. Employ t-SNE, a manifold learning algorithm, to reduce the dimensionality of the normalized dataset (Dataset I) while preserving the essential information.
ii. Output dataset_II, which is the resulting dataset with reduced dimensions suitable for further analysis.

2.1.2.1. Raw EEG-MI signal filtering. The initial step in preprocessing, EEG-MI signal filtering, is crucial for improving the signal-to-noise ratio of raw brainwaves and enhancing the relevant information within the EEG-MI signal. The EEG-MI signal is inherently susceptible to contamination from various sources, including body movements, eye blinks, facial muscle activity, and environmental artifacts such as electromagnetic fields generated by electrical devices. Therefore, it becomes imperative to employ filtering techniques to eliminate such noise and artifacts. This study used a fourth-order Butterworth filter for the EEG-MI signal filtering process. This filter was selected to eliminate signals contaminated by diverse noise sources effectively and to isolate rhythms within the frequency range of 8 Hz–30 Hz. This frequency range is particularly significant, as the EEG-MI method relies on detecting alpha rhythms (*8*–13 Hz) and beta rhythms (*14*–30 Hz) originating from the sensorimotor cortex [39]. Notably, there is a pronounced disparity in the size range between the two datasets. The application of the filtering process has inevitably led to considerable alterations in the size distribution, as is evident from the shift away from the original data distribution. These differences carry substantial implications, as they have the potential to introduce confounding factors and bias into subsequent analyses. Therefore, a meticulous assessment of the influence of data filtering on the ultimate outcomes of any analysis is crucial.

2.1.2.2. Raw EEG-MI segmentation. Segmentation is a critical process in signal analysis, and its performance significantly impacts the efficiency of subsequent steps, such as feature extraction and classification [39]. In this context, segmentation was employed to increase classification accuracy by selectively excluding periods or feature components within the EEG-MI signal. Importantly, each subject may exhibit varying MI signal powers at different times during the trial. This variability arises because individuals may commence (*or conclude*) their MI tasks at slightly different intervals owing to the inherent complexity of their brain processes. Consequently, to facilitate feature extraction and classification of brainwave data, the segmentation process was indispensable. This process involves partitioning the MI signal into distinct time frames or windows, each capturing specific patterns of neural activity. This study employed five distinct time-segment groups to partition the EEG-MI signal into various time windows or frames, each serving a particular purpose:

1. **Five-second time window group**: This group encompassed a time window ranging from 3 s to 8 s. It represents the entire signal duration from the initiation of the MI cue until the end of the signal.
2. **Four-second time window group**: Within this group, the time window was further subdivided into two subgroups on the basis of the time frames of 3–7 s and 4–8 s.
3. **Three-second time window group**: Similarly, this group was divided into three subgroups, each corresponding to distinct time windows: 3–6 s, 4–7 s, and 5–8 s.
4. **Two-second time window group**: Within this group, the time window was further divided into four subgroups, each associated with different time windows: 3–5 s, 4–6 s, 5–7 s, and 6–8 s.
5. **One-second time window group**: Last, the 1-s time window group was subdivided into five subgroups, each defined by specific time windows: 3–4 s, 4–5 s, 5–6 s, 6–7 s, and 7–8 s.

These distinct time-segment groups allowed us to analyze the EEG-MI signal with varying levels of granularity, owing to the intricacies of neural activity patterns during MI.

2.1.2.3. Time–Frequency domain feature extraction. The time–frequency domain represents a hybrid form of representation of brain signals [40]. This domain describes the neurophysiological characteristics of EEG-MI signals within both the time and frequency domains, thus offering a multilevel approach to signal analysis. The strength of time–frequency analysis lies in its ability to preserve frequency while still identifying temporal changes in the signals.

There are several methods of time–frequency analysis, including the autoregressive model (ARM), the short-time Fourier transform (STFT), the wavelet transform, and the discrete wavelet transform (DWT). Each of the methods has its benefits, but the DWT has been touted in the literature for its potential to represent EEG signals, particularly nonstationary signals. One of the primary advantages of the DWT is that it can provide excellent time–frequency localized analysis of a signal while remaining computationally efficient. Therefore, the DWT is a tool that researchers and practitioners would like to consider when dealing with complicated signals across disciplines [41].

DWT has become synonymous with analyzing brain signals in time–frequency domains on multiple scales, as it is isostochastic in its ability to localize a signal in time–frequency space. The DWT allows any general function to be represented as an infinite series of wavelets, making the wavelet transform a powerful method for spectral estimation.

In practical terms, DWT operates by dilating and shifting a specific function known as a mother wavelet. This process results in a signal represented as a linear combination of a set of wavelet functions. The mother wavelet (*polyharmonic*) generates these wavelets through shifting (*dilation*) and stretching (*translation*) operations along the time axis.

The application of the wavelet transform in biomedical signal processing has garnered significant attention because of its efficiency in diagnostic and pattern recognition [42]. The wavelet transform can be categorized into two main types: continuous wavelet transform (CWT) and DWT [43]. The CWT equation is defined as follows:

$$CWT(a,b) = \int_{-\infty}^{+\infty} x(t)\, \psi^{*}_{a,b}\,(t)dt \tag{1}$$

In this equation:

- where $x(t)$ represents the unprocessed EEG signal.
- a corresponds to the dilation factor.
- b signifies the translation factor.
- $\psi_{a,b}$ (t) denotes the complex conjugate and can be calculated as:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}}\, \psi\left(\frac{t-b}{a}\right) \tag{2}$$

Here, ψ(t) represents the wavelet function.

However, a significant limitation of the CWT is that its scaling parameter a and translation parameter b change continuously. Consequently, calculating wavelet coefficients for all available scales demands substantial computational effort but yields limited informative content [44]. The wavelet transform method can be likened to a mathematical microscope that decomposes a signal into multiple subsignals. Within this framework, the same signal can be represented across different frequency bands to capture frequency-specific information at appropriate time intervals.

In practice, the choice of decomposition level and the selection of mother wavelets are crucial factors when the DWT is employed for analyzing brain signals [45]. The frequency of interest determines the necessary decomposition levels to identify spatial patterns in brain signals. In this study, five decomposition levels were used in the wavelet transform, facilitating the extraction of alpha and beta band rhythms. Consequently, the alpha and beta bands were located at levels five and four, respectively. The power of these rhythms changed as the subjects simulated hand movements.

To reduce the high dimensionality of the feature space derived from wavelet coefficients, five statistical features were applied: maximum, minimum, median, mean, and standard deviation. These statistical measures effectively capture the essential characteristics of the original EEG-MI signal without introducing redundancy [46].

2.1.2.4. Merging and labeling process. Merging in data science refers to the process of combining two or more datasets into a single, unified dataset. The merging process depends on the standard columns in the dataset, which can be merged into a single file [47]. Labeling refers to the process of assigning a category or class to a set of data, resulting in a new column in a dataset [48]. Initially, the raw EEG-MI dataset consists of 30 features divided into 15 Excel sheets, each corresponding to a specific time-segment group representing the duration of hand movements (left or right). The dataset encompasses 15 time-segment groups distributed across various time intervals, including (3–4), (4–5), (5–6), (6–7), (7–8), (3–5), (4–6), (5–7), (6–8), (3–6), (4–7), (5–8), (3–7), (4–8), and (3–8). As part of the preprocessing steps, all 15 sheets are merged into a single file because of the similarity among the common columns. Second, another column labeled “time duration” was added to the time-segment group as an additional feature to the dataset, which was coded to values ranging from 1–15 for each time interval, resulting in a dataset with 31 features. For example, interval (3–4) was coded as 1, (4–5) was coded as 2, and so on. Additionally, a “movement” column was added as the target class, which includes two MI hand movement tasks: right and left. Right movement is coded as 1, and left movement is coded as 2. After the DWT features from all the time frames are merged into one file, the dataset is balanced, comprising a total of 21,600 records, with the target class split into 10,800 records for right movement (*label 1*) and 10,800 records for left movement (label 0). Balancing the dataset eliminates the need for augmentation techniques, as each movement class (left and right) is equally represented. Additionally, as a preprocessing step for the dataset, a “time duration” column is added, where each time duration is encoded as a categorical type by assigning a numeric value ranging from 1 to 15.

2.1.2.5. Normalization- produce dataset I. In this section, the objective is to generate the EEG-MI dataset by adopting many standard preprocessing steps. Normalization of dataset features is a technique employed to standardize the range of values for independent variables or features within a dataset. This standardization ensures that the various features in the dataset have almost identical scales, which enhances the accuracy and performance of the ML models used to train the dataset [49].

Min–max normalization, which transforms feature values to a range between 0 and 1, is one of the most famous methods for feature scaling. This can be achieved by subtracting the minimum feature value from each point value in the dataset and then dividing the result by the difference between the maximum and minimum values. The formula for normalizing a feature ’ X ′ is as follows:

$$Z = \frac{X - min(X)}{max(X) - min(X)} \tag{3}$$

where:

- X is the original value.
- Min(X) is the minimum value in the dataset.
- Max(X) represents the maximum value in the dataset.
- Z is the normalized value.

Visualizing the data before and after normalization is essential for

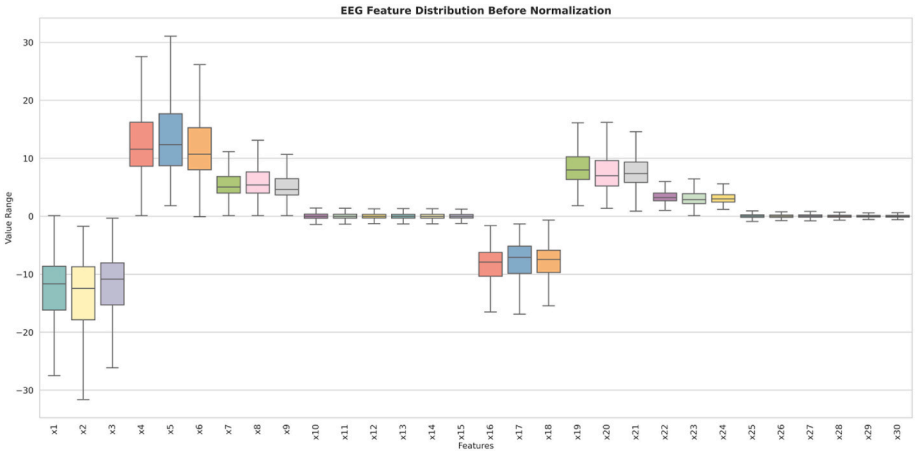


Fig. 4. EEG-MI dataset before normalization.

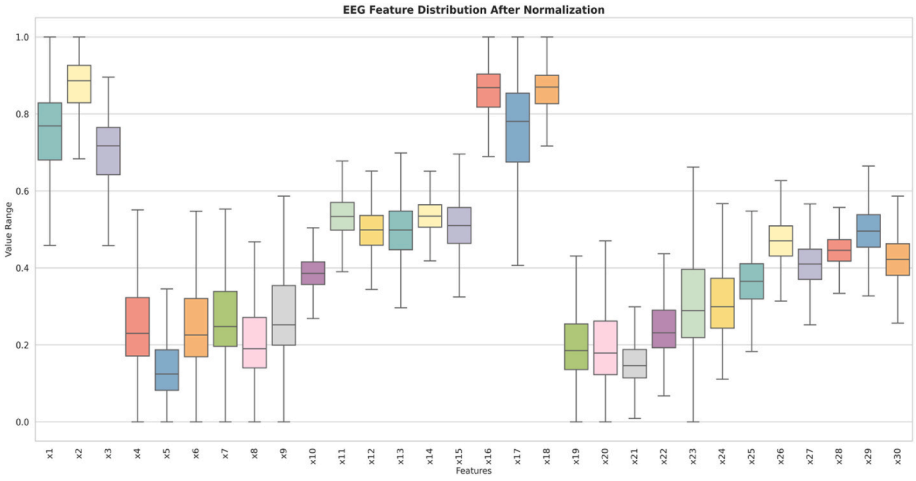


Fig. 5. EEG-MI dataset after normalization.

showing the effects of the normalization process on the EEG-MI data, as it helps in identifying abnormal patterns. Figs. 4 and 5 indicates visual representations of the EEG-MI dataset before and after normalization, grouped by movement class.

Fig. 4 shows the boxplot visualization for the statistical distribution of data values across different dataset values. The boxplot is a statistical tool that represents the distribution values of data inside each dataset feature. The box expresses the interquartile range (IQR), which indicates the values that fall between the first quartile (Q1 - 25 %) and the third quartile (Q3 - 75 %). The line inside the box represents the data median (which is the mean value in the feature), and the whiskers represent the smallest and largest values that fall within an acceptable range. Fig. 5 shows the boxplot visualization of the statistical distribution of each feature in the EEG_MI dataset after normalizing the dataset values to a range of 0–1. This normalization step is crucial for ensuring that features with differing units and scales are brought to a standard scale, facilitating easier comparison and analysis. In conclusion, at this stage, the normalized EEG-MI dataset represents the initial outcome dataset, referred to as "Dataset I." This dataset has undergone standard preprocessing. Dataset I employs multiple ML models to evaluate performance metrics via standard preprocessing. The subsequent section delves into further processing to facilitate fusion analysis and produce Dataset II.

2.1.2.6. Dataset II based on the feature fusion technique. The objective of this section is to produce the second EEG-MI dataset, referred to as Dataset II, which is constructed via feature fusion. Feature fusion is the combination of features or variables from multiple datasets into a unified set of features. This merging increases the performance of ML models by providing more informative and diverse input data. This technique finds applications in scenarios where data originate from multiple sources or sensors, such as in image processing, where data may come from various sensors or modalities [50]. Manifold learning is a data fusion technique used to convert high-dimensional data to lower-dimensional data by discovering the underlying data form. The manifold learning method fuses the dataset into a lower dimension by retaining significant data features, including geometric form, topology, and distance relationships. One of the most popular types is t-SNE, which is used for data visualization and dimensionality reduction. The t-SNE technique can effectively capture the local structure of data, visualize complex datasets and discover meaningful patterns. This ability is crucial for maintaining relationships between similar data points, which helps expose the underlying data structures that may not be apparent in high-dimensional spaces [51]. A simple comparison between the common types of data fusion is necessary to justify the reasons for preferring the t-SNE technique over other types, such as PCA and autoencoder. All these techniques are used for data fusion to reduce the dataset dimensionality, but each method has a different approach to implementation. t-SNE is a nonparametric method that preserves local structures and neighborhood distances, making it particularly well suited for visualization and clustering tasks. It also has the advantage of handling complex, nonlinear relationships effectively. Autoencoder

techniques, such as t-SNE, can also learn nonlinear embedding, but they focus more on minimizing reconstruction errors rather than preserving local structures. This makes them useful for feature learning but less effective for directly visualizing datasets.

Furthermore, they require significant mathematical effort because they rely on deep neural networks. t-SNE strikes a balance between the above methods by preserving local structures without requiring extensive training, making it a more suitable choice for visualizing high-dimensional data, such as EEG feature spaces. The t-SNE algorithm's objective is to minimize the discrepancy between these two probability distributions, employing a gradient descent method [52]. The equations of the t-SNE algorithm include probability distributions and a cost function that is minimized during optimization. The key equations in t-SNE include the high-dimensional probability distribution, the low-dimensional probability distribution, the cost function, and the optimization of the cost function [53], as follows:

i. High-dimensional probability distribution:

$$p(i,j) = \frac{\exp\left(-\frac{||xi-xj||^2}{2sigma^2}\right)}{sum\left(\exp\left(-\frac{||xi-xk||^2}{2sigma^2}\right)\right)}, \tag{4}$$

where:

xi and xj represent two high-dimensional data points.

Sigma is a parameter governing the distribution's scale, and the sum encompasses all other data points.

ii. Low-dimensional probability distribution:

$$q(i,j) = \frac{\left(1+||yi-yj||^2\right)^{-1}}{sum\left(\left(1+||yi-yk||^2\right)^{-1}\right)}, \tag{5}$$

where:

yi and yj denote the low-dimensional embeddings of data points xi and xj, respectively, and the sum spans all other data points.

iii. Cost function calculation:

$$C = KL(P \parallel Q) = sum\left(p(i,j)^* \log\left(\frac{p(i,j)}{q(i,j)}\right)\right) \tag{6}$$

where:

KL (P ||Q) signifies the Kullback–Leibler divergence between the high-dimensional probability distribution P and the low-dimensional probability distribution Q, and the sum encompasses all pairs of data points.

iv. Optimization of the cost function:

The t-SNE algorithm employs a gradient descent method to minimize the cost function C with respect to the low-dimensional embeddings *yi*. The gradient of C concerning *yi* is expressed as:

$$\frac{dC}{dyi} = 2 * sum(p(i,j) - q(i,j))^* (yi - yj)^* \left(1+||yi-yj||^2\right)^{-1} \tag{7}$$

The optimization process iteratively updates the low-dimensional embedding *yi* via the gradient descent rule until the cost function converges to a minimum. In this study, (Dataset I) is normalized to scale the dataset to values between 0 and 1 (as previously shown in Fig. 5) and is finally processed through manifold learning via the t-SNE method, resulting in 9 fused features, as illustrated in Fig. 6.

2.2. Phase 2: development of multiple ML models

This phase describes the process of constructing multiple ML models for a robotic hand control system while investigating the effectiveness of adversarial and nonadversarial attack strategies on two preprocessed datasets that underpin a robotic hand control model. Dataset I undergoes standard preprocessing, whereas Dataset II adopts feature fusion preprocessing. These ML models are constructed via nine distinct ML algorithms, namely, kNN, DT, SGD, SVM, RF, GB, NN, AdaBoost, and LR. These algorithms were chosen for their ease of understanding and the ease with which their results can be explained. They are less complex and require fewer parameters to tune, relying on well-established statistical theories, which also enables their application to a wide range of problems. In addition, these ML methods have been successfully applied in various fields, particularly in the medical field, and their effectiveness provides substantial grounds for their continued use.

Each constructed model subsequently undergoes a quality assessment through evaluation metrics. The nonadversarial model is developed through a process that encompasses training, testing, and validation of the dataset without any adversarial attacks. In contrast, models intended to resist adversarial attacks are referred to as adversarial ML models. These models are trained with adversarial examples to increase their resistance to these attacks. An adversarial example refers to a modified input that induces the model to produce inaccurate predictions. Various methods for launching adversarial attacks, including gradient-based attacks, optimization-based attacks, and black-box attacks, have been discussed [54–56]. Gradient-based attacks leverage the model's gradients to determine the necessary input modifications that

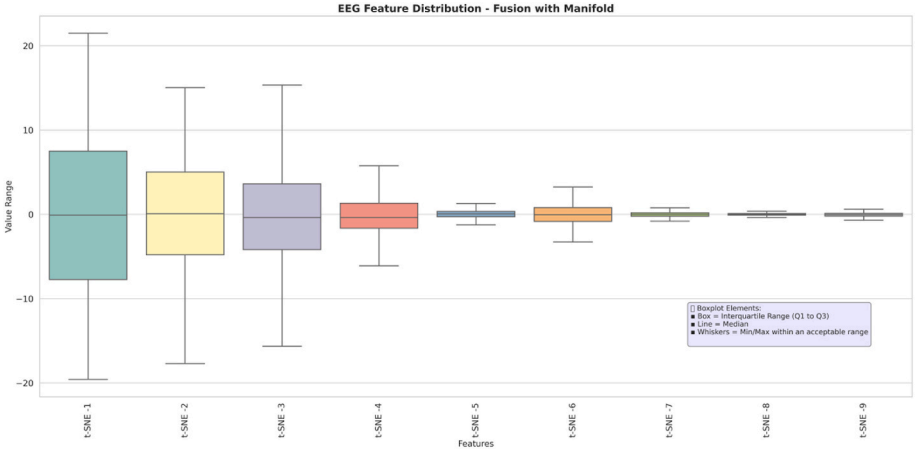


Fig. 6. EEG_MI dataset after feature fusion via t-SNE.

disrupt accurate predictions.

In contrast, optimization-based attacks employ optimization techniques to identify inputs that cause erroneous outputs. Black-box adversarial attacks generate adversarial examples by approximating the target model, where the attacker does not need access to the internal parameters or architecture of the attacked ML model. Instead, the attacker can only query the ML model and observe its outputs. Evaluating performance is crucial for gauging the effectiveness of the robotic hand control system in achieving its objectives. Performance evaluation measures are employed to assess the developed models for robotic hand control via the tested datasets. Selecting the appropriate metrics is a pivotal step. For this purpose, nine criteria are employed, with a focus on prominent metrics such as the area under the curve (AUC), Acc, F1 score, precision, recall, log loss, specificity, training time, and testing time. Table 1 provides detailed formulations for these metrics:

In this study, a k-fold cross-validation strategy was employed in the model evaluation process by using unseen MI-EEG inputs. In the k-fold process, the procedure was iteratively performed k times, dividing the dataset into k subgroups of data. k-1 subgroups were used for training, and the remaining subgroups were used for testing. The folds are divided randomly to ensure that the entire dataset is involved in each fold.

2.3. Phase 3: evaluation and benchmarking framework

This phase employs a multiperspective approach to construct two decision matrices that facilitate the selection of the optimal model for robotic hand control. To do this, I specifically utilized the dataset to evaluate models without the integration of feature fusion. Moreover, the second dataset (II) was designed to assess models via feature fusion techniques. Following the formulation of these decision matrices, a series of evaluations and benchmarking processes were methodically conducted. These assessments were performed with a focus on objectivity, utilizing the FDOSM method as the basis for assessment. The primary objective of these measures was to establish a comprehensive and evidence-based foundation upon which to identify the most suitable model for the intended application.

2.3.1. Multiperspective decision matrix (MPDM)

A multiperspective approach is a systematic methodology that integrates diverse viewpoints or perspectives to address complex problems or research questions effectively. In the context of ML, adopting a multiperspective approach can yield a more holistic and nuanced understanding of the data or problem, potentially leading to enhanced solutions and insights. Selecting the optimal ML model for robotic hand control is a critical task that necessitates the consideration of both adversarial and nonadversarial factors. Each of these factors inherently involves a multiperspective approach, and the inclusion of both standard and feature fusion preprocessing techniques will add further complexity. To navigate this intricate landscape, the development of an MPDM becomes essential. The MPDM accommodates both adversarially trained and nonadversarially trained models, ensuring a comprehensive analysis of the two preprocessed datasets, as outlined in Table 2. Consequently, a meticulous evaluation of the various models is imperative. This evaluation aims to discern the respective strengths and weaknesses of each model, facilitating the selection of the most trustworthy and explainable model for the intended robotic hand control system.

2.3.2. FDOSM method

FDOSM, an MCDM method, plays a significant role in selecting the optimal ML model by considering multiple perspectives and criteria. It provides a mathematical framework for addressing MCDM challenges, encompassing both individual and group decision-making contexts. FDOSM consists of three primary block units: the data input unit, the data transformation unit, and the data processing unit [60]. The framework for group decision-making within FDOSM comprises two essential steps: external and internal aggregation. These FDOSM steps can be succinctly described as follows (see Fig. 7):

Table 1
The employed performance metrics.

Evaluation metrics	Mathematical Equation	Explanation
Acc	$CA = \frac{TP + TN}{TP + FP + FN + TN}$	In the context of constructing multiple ML models for a robotic hand control system, Acc measures the ratio of correctly classified hand movement instances by the ML models to the total number of cases in the dataset.
Precision	$\text{Precision} = \frac{TP}{TP + FP}$	Precision quantifies the number of correctly classified hand movement instances by the ML models among all the cases that the models classified as hand movements. It helps assess the models' ability to avoid false positive predictions, which is crucial in robotic hand control.
Recall (Sensitivity)	$Recall = \frac{TP}{TP + FN}$	Recall measures the number of correctly classified positive hand movement instances by the ML models. It focuses on the models' ability to identify all actual positive hand movements, which is essential in robotic control applications.
F1-score	$F1 - \text{score} = \frac{2*TP}{2*TP + FP + FN}$	The F1 score is a combined measure of precision and sensitivity tailored for the robotic hand control models. It provides a balanced assessment of the models' performance by considering both their ability to minimize false positives and false negatives during hand movement predictions.
Specificity	$Specificity = \frac{TN}{TN + FP}$	Specificity evaluates the models' ability to correctly classify negative instances (nonhand movements) and identify true negatives for each category. It helps assess the models' ability to distinguish between hand movements and other activities.
AUC	$\text{AUC} = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right)$	AUC measures the models' ability to discriminate between positive (hand movement) and negative (nonhand movement) instances across different thresholds. It provides insights into the models' overall discriminatory power in the context of robotic hand control.
Log Loss	$\text{LogLoss} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log_e(\hat{y}_i) + (1 - y_i)\log_e(1 - \hat{y}_i)\right]$ n: number of samples, y_i: is the true label of the i-th instance, where $y_i \in \{0,1\}$ $\hat{y}_i$ is the predicted probability	Log loss assesses the models' performance in predicting probabilities of hand movements. It quantifies how well the models' predicted probabilities align with the actual outcomes of hand movements, which is crucial for reliable control in robotic applications.

TP: True Positive, TN: True Negative, FP: False Positive, TN: True Negative, FN: False Negative.

Table 2
Development of MPDM.

Alternatives (Multiple ML Models based on Robotic Hand Control)	Non-Adversarial Perspective Criteria									Adversarial Perspective Criteria								
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C1	C2	C3	C4	C5	C6	C7	C8	C9
A1	C1 – A1	C2 – A1	C3 – A1	C4 – A1	C5 – A1	C6 – A1	C7 – A1	C8 – A1	C9 – A1	C1 – A1	C2 – A1	C3 – A1	C4 – A1	C5 – A1	C6 – A1	C7 – A1	C8 – A1	C9 – A1
A2	C1 – A2	C2 – A2	C3 – A2	C4 – A2	C5 – A2	C6 – A2	C7 – A2	C8 – A2	C9 – A2	C1 – A2	C2 – A2	C3 – A2	C4 – A2	C5 – A2	C6 – A2	C7 – A2	C8 – A2	C9 – A2
….	……	……	……	……	……	……	……	……	……	……	……	……	……	……	……	……	……	……
A9	C1 – A9	C2 – A9	C3 – A9	C4 – A9	C5 – A9	C6 – A9	C7 – A9	C8 – A9	C9 – A9	C1 – A9	C2 – A9	C3 – A9	C4 – A9	C5 – A9	C6 – A9	C7 – A9	C8 – A9	C9 – A9

This Fig. 7 illustrates the comparison process of the FDOSM scores of various ML models, highlighting their relative performance under the same evaluation criterion. The chart also demonstrates the proximity of each model's score to the ideal solution within the same criterion, where lower scores indicate better alignment with the ideal model performance.

Stage 1- Data input unit: In the data input unit of the proposed MCDM method, m options are assigned to each MCDM problem, denoted as $A1, A2, \ldots Am$, representing the multiple ML models under consideration. Additionally, $C1, C2, \ldots Cn$ are included, representing n sets of decision perspective criteria related to the nine evaluation metrics. The outputs of this unit are captured by decision makers (DMs), and the decision matrix is subsequently transformed into an opinion matrix.

Stage 2- Data Transformation Unit: In FDOSM, the data transformation unit involves the following steps:

1. A three-parameter optimal solution is identified on the basis of the outputs generated by the DMs in the previous block. These three parameters include the minimum, maximum, and critical values.
   - o **Minimum value criterion**: This criterion is employed for cost-related criteria, where the option with the lowest value is considered the best.
   - o **Maximum value criterion**: In contrast, the maximum value criterion is used for benefit-related criteria, where the option with the highest value is regarded as the optimal choice.
   - o **Critical value**: In certain situations, particularly when the ideal response is neither minimal nor maximal (as in the case of blood pressure), the critical value philosophy is employed. To select the best ML model, it is essential to specify a critical value for the evaluation criteria. For instance, in the DM, criteria such as C1 (training time), C2 (testing time), and C8 (log loss) are treated as costs, whereas the other criteria are considered benefits.

In this step, the process of selecting the optimal model within each criterion result is described by the following equation:

$$A^{*} = \left\{ \left[\left(\max_{i} v_{ij} \middle| j \in J \right), \left(\min_{i} v_{ij} \middle| j \in J \right), \left(Op_{ij} \in I.J \right) | i = 1.2.3.\ldots.m \right] \right\} \tag{14}$$

where:

" min" represents the optimal solution for the cost criterion,

" max" represents the optimal value for the benefit criteria, and

" Op_{ij}" indicates the critical value where the optimal value falls between " min" and " max".

2. **Opinion Scores via Five-Scale Linguistic Terms:** This involves comparing the optimal solution against alternative values on the basis of the criteria. This step implicitly provides a mechanism for weighting evaluation criteria. Decision-makers subjectively assess the significance of the differences between the ideal solution and alternatives. Five scales are used to compare linguistic expressions, ranging from "no difference" to "large difference" (Table 3). Eq. (15) facilitates the comparison between the optimal solution and other values. The output of this block is a linguistic term opinion matrix, which is then prepared for fuzzy membership-based translation into fuzzy numbers.

$$Op_{Lang} = \left\{ \left(\left(\widetilde{v}_{ij} \otimes v_{ij} \middle| j \in J \right) . | i = 1.2.3.\ldots.m \right) \right\} \tag{15}$$

At this step, a panel of three ML specialists, each with more than five years of expertise in robotics and control, was consulted.

Stage 3- Data Processing Unit: The opinion matrix, which is the output of the transformation unit, serves as the basis for the final block's operations. This last block initiates by transforming the opinion matrix into a fuzzy opinion decision matrix via triangular fuzzy numbers

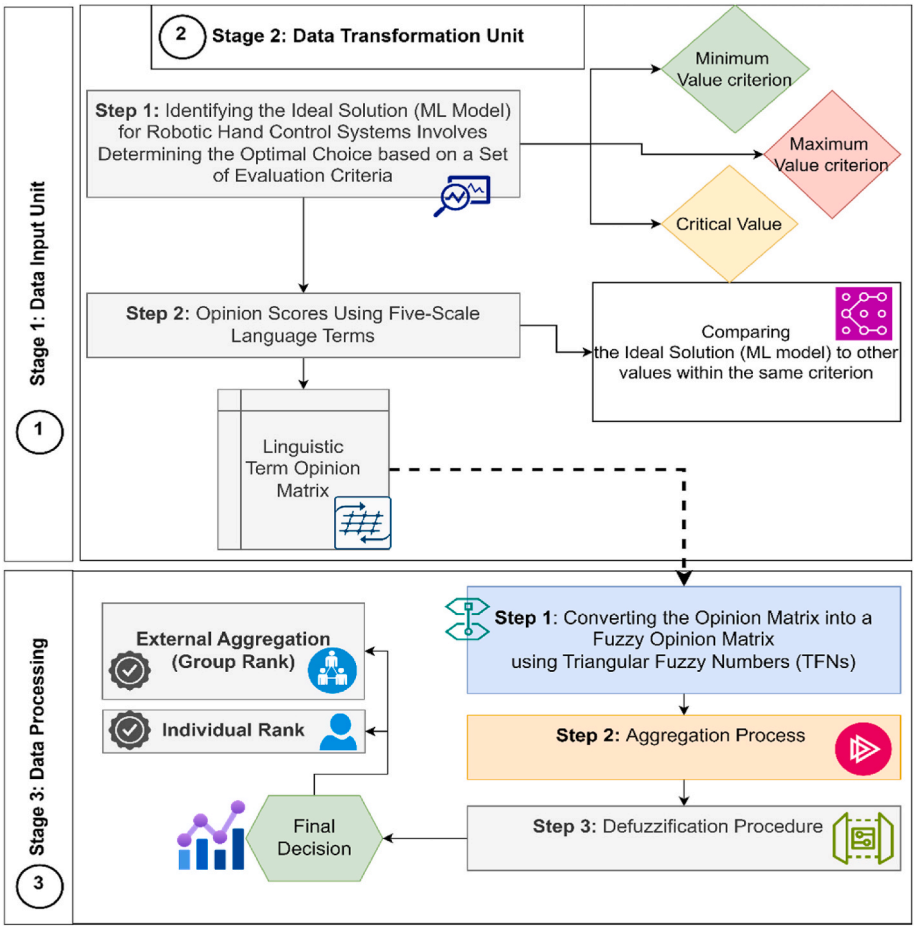


Fig. 7. FDOSM scores across the ML models (lower scores indicate better performance).

Table 3
Linguistic terms and their equivalent TFNs.

Linguistic terms	TFNs
No difference	(0.00,0.10,0.30)
Slight difference	(0.10,0.30,0.50)
Difference	(0.30,0.50,0.75)
Big difference	(0.50,0.75,0.90)
Enormous difference	(0.75,0.90,1.00)

(TFNs). A direct aggregation operator is subsequently applied, specifically, the arithmetic mean. The steps for this level are outlined as follows:

1. The application of triangular fuzzy membership to the opinion matrix results in the generation of a fuzzy decision matrix, which is part of the fuzzification procedure. TFNs are used to convert the terms of the opinion matrix, with the TFN function determining their membership.

$$\mu A(x) = \begin{cases} 0 \text{ if } x < a \\ \dfrac{x-a}{b-a} \text{ if } a \le x \le b \\ \dfrac{c-x}{c-b} \text{ if } b \le x \le c \\ 0 \text{ if } x > c \end{cases} \quad \text{where } a \le b \le c.$$

$\tilde{x} = (a1, b1, c1)$ and $\tilde{y} = (a2, b2, c2)$ are two TFNs, both of which are nonnegative, and α is a positive real number. The arithmetic operations are defined as follows, following the extension principle:

$$\tilde{x} + \tilde{y} = (a1 + a2, b1 + b2, c1 + c2),$$

$$\tilde{x} - \tilde{y} = (a1 - c2, b1 - b2, c1 - a2),$$

$$\alpha\tilde{x} = (\alpha a1, \alpha b1, \alpha c1),$$

$$\tilde{x}^{-1} \cong (1/c1, 1/b1, 1/a1),$$

$$\tilde{x} \times \tilde{y} \cong (a1a2, b1b2, c1c2),$$

$$\tilde{x} / \tilde{y} \cong (a1/c2, b1/b2, c1/a2).$$

Table 3 presents the TFNs for all the linguistic terms.

2. An aggregation operator, specifically the arithmetic mean, is used to combine the results obtained in the previous step for each option. After the fuzzy decision matrix is constructed, this aggregation process is applied to determine the optimal choice.

$$A_m(x) = \left(\sum \left(a_f + a_m + a_i\right)\left(b_f + b_m + b_i\right)\left(c_f + c_m + c_i\right) \right) \Big/ n \quad (16)$$

3. The defuzzification procedure is carried out via the centroid approach, as described by Eq. (17).

$$defuzzification = \frac{a+b+c}{3} \quad (17)$$

The best-ranking order is correlated with the lowest mean value.

Individual Rank: These rank results are established by processing the opinion matrix of a single DM independently. The fuzzy opinion matrix obtained from the expert undergoes aggregation and defuzzification, resulting in a specific ranking of the ML models on the basis of the DM's criteria and preferences.

External Group Aggregation: In external aggregation, fuzzy opinion matrices from multiple DMs are independently processed according to the procedures outlined in the processing unit. The outputs of these decision matrices are then aggregated via the arithmetic mean to establish the final group decision. In this scenario, expert opinions are combined after the final ranking is determined.

3. Results and discussion

The subsequent sections present an analysis of the outcomes resulting from the application of the elements of the methodology's phases, as previously described. Each of the included phases is discussed individually, illustrating how it relates to the overall research objectives and takes into account some judgment as to the appropriateness of the methodology.

3.1. Results of multiple ML models

In this section, the results of using multiple ML models for both the fusion and standard approaches are presented, and these results are analyzed to highlight the strengths, weaknesses, and factors that affect the performance of each approach to determine the optimal approach for the robotic hand control system on the basis of the EEG sensor.

3.1.1. ML models based on standard preprocessing

This section presents the evaluation results of multiple ML models applied to the normalized Dataset I, which is the EEG-based robotic hand control system. Tables 4 and 5 present the evaluation results of the adversarial and nonadversarial situations:

In Table 4, the ML model results show that the GB algorithm achieved the highest AUC score of 0.803, followed by RF with 0.788 and kNN and LR, both with AUC scores of 0.755. In contrast, the SVM algorithm exhibited the lowest AUC at 0.536. In terms of Acc, the GB algorithm achieves the highest accuracy value of 0.719, which is close to that of the RF algorithm (0.714) and the LR algorithm (0.691). The SVM, on the other hand, recorded the lowest Acc of 0.527. For the precision, recall, and specificity metrics, GB and RF secured the highest values, each with a precision of 0.714. However, although GB performed well in these metrics, it presented a longer training time, taking 14.732 units of time, which is relatively higher than those of the other models.

Additionally, both GB and RF presented the highest log loss values, with values of 0.342 for GB and 0.312 for RF. The results from the ML models presented in Table 4 are diverse, and determining the best model on the basis solely of Dataset I may not provide a clear picture. It is essential to consider adversarial attacks on these models, as their performance in such scenarios can significantly impact the decision-making process. In terms of accuracy and AUC under nonadversarial conditions, SVM performs poorly, whereas kNN, AdaBoost, and DT perform moderately well. SGD achieves a balanced accuracy, precision, and recall of 0.675. Both the LR and NN methods perform well, with the RF and GB methods achieving the highest performance among all the models. SGD, LR, and kNN are the most efficient methods, given their short training and testing times. RF and AdaBoost have moderate Acc values, whereas NN, SVM, and GB have lower Acc values. Notably, DTs have a longer training time but very rapid testing.

In Table 5, the results of the ML models under adversarial conditions indicate that the GB algorithm achieves the highest AUC of 0.781, followed by RF with 0.773 and NN with 0.753. Conversely, the SVM algorithm exhibited the lowest AUC at 0.528. With respect to Acc, the RF algorithm achieves the highest value of 0.702, closely followed by GB with 0.698 and LR with 0.680. The SVM yielded the lowest Acc of 0.509. In terms of precision, RF obtained the highest value at 0.702, closely followed by GB, with a value of 0.701. For recall, GB secured the highest value of 0.716, whereas SGD achieved the best LogLoss value of 0.215 among the models. GB and RF also demonstrated the highest specificity values at 0.718 and 0.710, respectively. In terms of the Acc and AUC, GB and RF exhibit the best performance, whereas LR and NN perform relatively well.

kNN, DT, and AdaBoost achieve moderate performance, and SGD achieves balanced performance. SVM has low performance. In terms of Acc, SGD and LR are the most efficient, whereas SVM and GB are less efficient. AdaBoost has a moderate Acc. Tables 4 and 5 present the visualizations of the developed ML models, which are based on the normalized Dataset I, to facilitate comparisons between nonadversarial and adversarial attacks for each evaluation metric. This visualization provides a comprehensive overview of how models perform under different conditions. The comparisons between Tables 4 and 5 clearly demonstrate the practical impact of adversarial attacks: a consistent drop in key performance metrics across models, accompanied by increased training times. This highlights the importance of considering adversarial robustness in model selection for EEG-based robotic hand control systems, as adversarial conditions can notably degrade model efficacy.

In summary, in terms of performance metrics, nonadversarial ML models outperform those affected by adversarial attacks. Nevertheless, the conflicts observed in the evaluation results reveal the critical need for a robust decision-making approach when selecting the most suitable algorithms for EEG-based robotic hand control systems. This selection ensures high efficiency and reliability in performance metrics while considering both nonadversarial and adversarial attack situations.

3.1.2. ML models based on feature fusion preprocessing

Tables 6 and 7 provide a comprehensive performance comparison between nonadversarial and adversarial methods, evaluated within the context of the fusion-based approach applied to Dataset II. The results presented in these tables aim to illustrate the relative effectiveness, strengths, and limitations of each method, offering a detailed assessment of how adversarial strategies impact the overall performance when integrated with the fusion framework (Dataset II).

In Table 6, the GB algorithm achieves outstanding results, with the highest AUC, Acc, F1 score, precision, and specificity, with scores of

Table 4
Evaluation of the ML models on the normalized dataset I (nonadversarial situations).

Model	Train time	Test time	AUC	Acc	F1	Precision	Recall	LogLoss	Specificity
kNN	2.568	0.384	0.733	0.670	0.670	0.670	0.670	0.685	0.670
DT	5.721	0.043	0.651	0.689	0.688	0.689	0.689	0.584	0.689
SGD	**1.423**	0.107	0.675	0.675	0.675	0.676	0.675	0.635	0.675
SVM	8.315	0.595	0.536	0.527	0.527	0.527	0.527	0.882	0.527
RF	2.679	0.107	0.788	0.714	0.714	0.714	0.714	**0.312**	0.714
GB	14.732	0.136	**0.803**	**0.719**	**0.719**	**0.719**	**0.719**	0.342	**0.719**
NN	11.086	0.094	0.755	0.690	0.690	0.690	0.690	0.596	0.690
AdaBoost	3.791	0.061	0.686	0.686	0.686	0.686	0.686	0.687	0.686
LR	2.190	**0.022**	0.755	0.691	0.691	0.691	0.691	0.599	0.691

Table 5
Evaluation of the ML models on the normalized dataset I (adversarial conditions).

Model	Train time	Test time	AUC	Acc	F1	Precision	Recall	LogLoss	Specificity
kNN	4.322	1.658	0.705	0.663	0.662	0.652	0.665	0.456	0.642
DT	9.215	0.700	0.632	0.664	0.678	0.675	0.673	0.355	0.676
SGD	**2.422**	**0.641**	0.672	0.670	0.664	0.672	0.650	**0.215**	0.646
SVM	12.594	1.524	0.528	0.509	0.511	0.526	0.509	0.654	0.524
RF	3.218	1.065	0.773	**0.702**	**0.709**	**0.702**	0.693	0.246	0.710
GB	19.155	0.749	**0.781**	0.698	0.695	0.701	**0.716**	0.236	**0.718**
NN	11.846	1.065	0.753	0.668	0.663	0.676	0.678	0.549	0.689
AdaBoost	4.198	0.805	0.674	0.669	0.675	0.682	0.657	0.846	0.673
LR	2.538	0.937	0.741	0.680	0.673	0.670	0.668	0.550	0.669

Table 6
Results of the ML models considering the nonadversarial method on feature fusion Dataset II.

Model	Train time	Test time	AUC	Acc	F1	Precision	Recall	LogLoss	Specificity
kNN	10.706	0.054	0.821	0.756	0.756	0.756	0.756	0.647	0.756
DT	7.331	0.016	0.708	0.712	0.712	0.713	0.712	0.489	0.712
SGD	0.758	0.072	0.725	0.725	0.725	0.725	0.725	**0.315**	0.725
SVM	3.091	0.013	0.592	0.574	0.521	0.583	0.574	0.899	0.574
RF	1.886	0.025	0.851	0.777	0.777	0.777	0.777	0.425	0.777
GB	1.285	0.098	**0.862**	**0.779**	**0.779**	**0.779**	**0.779**	0.452	**0.779**
NN	8.687	0.083	0.815	0.750	0.750	0.750	0.750	0.684	0.750
AdaBoost	3.672	0.042	0.746	0.746	0.746	0.746	0.746	0.549	0.746
LR	**0.159**	**0.009**	0.743	0.689	0.688	0.689	0.689	0.755	0.689

Table 7
Results of the ML models when the adversarial mechanism based on feature fusion is considered on Dataset II.

Model	Train time	Test time	AUC	Acc	F1	Precision	Recall	LogLoss	Specificity
kNN	1.120	0.659	0.816	0.737	0.751	0.755	0.738	0.237	0.731
DT	8.111	0.264	0.705	0.699	0.694	0.687	0.707	0.807	0.707
SGD	1.113	0.510	0.721	0.725	0.701	0.707	0.721	0.326	0.701
SVM	3.811	0.483	0.571	0.545	0.509	0.574	0.557	0.929	0.564
RF	2.564	0.384	**0.846**	0.753	**0.768**	0.760	**0.768**	**0.235**	**0.767**
GB	2.065	0.924	0.845	**0.777**	0.764	**0.762**	0.766	0.265	0.757
NN	3.055	0.904	0.813	0.727	0.727	0.723	0.722	0.324	0.732
AdaBoost	4.635	**0.086**	0.742	0.733	0.746	0.732	0.735	0.302	0.734
LR	**1.032**	0.980	0.715	0.685	0.670	0.676	0.661	0.586	0.678

0.862, 0.779, 0.779, 0.779, and 0.779, respectively. However, it falls short in the LogLoss metric, with a value of 0.452. On the other hand, the SGD algorithm achieves the best log-loss performance. In terms of training time, kNN exhibited the longest training process, consuming 10.706 units of time, which stands out compared with the other algorithms. Conversely, the testing times across all the ML models were uniformly efficient, with LR boasting the shortest testing time at 0.009. These results highlight the trade-offs and challenges in selecting the optimal ML model. While GB excels in several metrics, its log loss performance raises concerns, and kNN's extended training time is a drawback. Deciding on the best ML model is particularly challenging when considering adversarial attacks on the dataset, as these conflicting outcomes underscore the importance of evaluating both adversarial and nonadversarial scenarios.

The results reported in Table 7 reveal notable differences compared with those in Table 6, which can be attributed primarily to the common influence of adversarial ML models. These adversarial models tend to decrease performance metrics because of the computational overhead they entail. Consequently, it becomes evident that the nonadversarial approach consistently outperforms the adversarial approach across the majority of models. Nevertheless, choosing the optimal model should encompass both nonadversarial and adversarial perspectives for a comprehensive evaluation. Figures A1 and A2 (Appendix) visually illustrate these comparisons between the normal and adversarial attack scenarios developed via Datasets I and II.

These results demonstrate that adversarial attacks generally cause a decline in performance metrics, such as Acc and AUC, across most models; however, the magnitude of impact varies. For example, the SVM model's Acc decreases by approximately 5.3 % (from 0.574–0.545), indicating a greater susceptibility than the GB and RF models do, whose Acc changes by less than 0.3 % (GB: 0.779–0.777; RF: 0.777–0.753). This practical comparison clearly highlights the robustness of ensemble methods, such as GB and RF, against adversarial attacks, whereas models such as SVM and kNN experience more pronounced degradation. The relatively stable Acc and AUC values of GB (AUC: 0.862–0.845) and RF (AUC: 0.851–0.846) under adversarial conditions suggest that feature fusion preprocessing enhances model resilience by providing richer data representations that help mitigate the effects of adversarial attacks.

Moreover, when the nonadversarial results of feature fusion dataset II (see Table 6) are compared with those of standard preprocessing dataset II (see Table 4), it becomes evident that the ML models achieve higher performance values with feature fusion preprocessing. In these contexts, the incorporation of feature fusion techniques contributed to enhanced model capabilities, highlighting the potential of fusion preprocessing for optimizing performance. Additionally, in the case of the adversarial results for feature fusion dataset II (see Table 7) compared with those for standard preprocessing dataset I (see Table 5), the results also highlight the promise of fusion preprocessing in addressing adversarial challenges. While adversarial attacks typically lead to performance degradation, the models preprocessed with feature fusion exhibited resilience and maintained competitive performance metrics, emphasizing the robustness of this approach in adversarial scenarios. This finding indicates that feature fusion preprocessing can be

particularly valuable in scenarios where adversarial threats are a concern, such as EEG-based robotic hand control systems.

Feature fusion preprocessing not only reduces the dataset dimension but also demonstrates superiority in enhancing the performance and efficiency of ML models against adversarial attacks. This is clearly observed in the evaluation values and their differences between feature fusion preprocessing and standard preprocessing under nonadversarial and adversarial attack conditions. Analysis and comparison of the ML models developed on the basis of both datasets reveal clear trade-offs and conflicts in multicriteria evaluation metrics. These conflicts arise from the presence of both benefit and cost criteria. The benefit criterion implies that higher values are better, whereas the cost criterion suggests that lower values are preferable. This inherent conflict makes it challenging to determine the superior model, as there is often a trade-off between different criteria. To navigate these complexities and select the optimal model, an MCDM process is employed. The following section introduces an evaluation and benchmarking framework designed to address these issues, focusing on resolving conflicts, managing trade-offs, and establishing the significance of evaluation metrics.

3.2. Evaluation and benchmarking results

This section addresses the key issues of conflict, trade-offs, and metric importance that arise when the best ML model is selected. To address these challenges, we employ a comprehensive evaluation and benchmarking framework that combines the MPDM and FDOSM methodologies. First, MPDM is utilized to evaluate the ML models developed on Dataset I, which undergoes standard preprocessing. This evaluation considers both nonadversarial and adversarial attack perspectives. Additionally, MPDM is applied to assess the ML models constructed via Dataset II, which undergoes feature fusion preprocessing. Furthermore, the integration of FDOSM with MPDM enables us to make informed decisions about selecting the optimal ML model. As mentioned earlier, the FDOSM process involves inputs from three experts, contributing to a more comprehensive and robust model selection process. The explanation in this paragraph outlines the process for both datasets.

Initially, Eq. (14) is employed to select the optimal model within each criterion result. For example, experts select the best model result for Acc. Additionally, during the application of Stage 1/Data Transformation Unit, values for “min,” “max,” and " Op_{ij}" must be set. Subsequently, Stage 2 within the same unit involves comparing the optimal ML model solution against other values within the same criterion (metric) via Eq. (15). In the application of Stage 1/Data Transformation Unit, linguistic terms and their equivalent TFNs from Table 3 are utilized. At this juncture, Eqs. (16) and (17) are applied during the processes of Stages 2 and 3 of the data transformation unit to derive the final ranking results from the input of three experts, denoted as individual benchmarking. To determine the most crucial criteria for evaluating robotic hand control models, consultations were conducted with three data science experts. These experts were selected because of their extensive experience in the field of data science. They evaluated the developed ML models on both datasets. The results of the evaluation and benchmarking are presented in the subsections below, with each section presenting its results according to the preprocessing approach used for its respective dataset.

3.2.1. Benchmarking ML models using Dataset I (standard preprocessing)

Fundamentally, Table 8 presents the results of the evaluation and benchmarking carried out by each expert via the standard preprocessing approach, considering two perspectives: nonadversarial (see Table 4) and adversarial attacks (see Table 5). The performance score represents the ranking of each ML model, with a lower score indicating better performance.

The experts completely agreed that the best model is the RF. However, there was a disagreement regarding the ranking of the GB model. The second and third experts ranked GB as the second-best model, whereas the first expert ranked it as the sixth-best model. On the other hand, the SVM model performed the worst among all the models evaluated by the experts, ranking ninth. These results underscore the importance of considering multiple expert opinions and the necessity of a rigorous evaluation process when selecting the optimal model for robotic hand control. Table 9 provides an overview of the results obtained for the evaluation benchmarking group. The three experts employed an external grouping method, utilizing the arithmetic mean.

The results show that the RF model achieved the best performance score (0.18241), indicating its superior performance among all the models. Conversely, the SVM model obtained the worst performance score (0.69475), indicating that it performed worse than the other models did. Interestingly, the GB model received the second-best performance score, with a value of 0.31327, highlighting its potential for achieving high-performance results. However, the DT model received the second-worst model performance score, with a value of 0.42407, suggesting that its performance requires improvement.

3.2.2. Benchmarking ML models using Dataset II (feature fusion preprocessing)

Table 10 presents the results of the evaluation and benchmarking carried out by each expert via the feature fusion approach for Dataset II. The performance scores represent the rankings of each ML model, with lower scores indicating better performance.

Table 9
Group benchmarking results for the ML models developed on Dataset I.

ML Models	Group Performance Score	Rank
kNN	0.42099	6
DT	0.42407	8
SGD	0.39105	5
SVM	0.69475	9
RF	**0.18241**	**1**
GB	0.31327	2
NN	0.38673	4
AdaBoost	0.42377	7
LR	0.34136	3

Table 8
Individual benchmarking results for the ML models developed on Dataset I.

ML Models	Expert 1		Expert 2		Expert 3	
	Performance Score	Rank	Performance Score	Rank	Performance Score	Rank
kNN	0.37963	5	0.42778	6	0.45556	8
DT	0.41759	8	0.43796	7	0.41667	7
SGD	0.37130	4	0.40926	5	0.39259	4
SVM	0.52778	9	0.74907	9	0.80741	9
RF	**0.16389**	**1**	**0.17315**	**1**	**0.21019**	**1**
GB	0.40093	6	0.23889	2	0.30000	2
NN	0.34444	3	0.40370	4	0.41204	6
AdaBoost	0.41574	7	0.44907	8	0.40648	5
LR	0.33611	2	0.35926	3	0.32870	3

Table 10
Individual benchmarking results for the ML models developed on Dataset II.

ML Models	Expert 1		Expert 2		Expert 3	
	Performance Score	Rank	Performance Score	Rank	Performance Score	Rank
KNN	0.39074	4	0.17315	4	0.34630	3
DT	0.63241	7	0.29444	7	0.59722	8
SGD	0.47685	6	0.17037	3	0.41111	5
SVM	0.72130	9	0.71667	9	0.78056	9
RF	**0.23796**	**1**	**0.14259**	**1**	0.26852	2
GB	0.31204	2	**0.14259**	**1**	**0.25741**	**1**
NN	0.44537	5	0.18241	6	0.43704	6
AdaBoost	0.35370	3	0.17963	5	0.37315	4
LR	0.64537	8	0.41852	8	0.58333	7

The first and second experts agreed that the RF model is the best, whereas the third expert agreed that it is the second-best model. Both the second and third experts decided that the GB is the best model. In contrast, the SVM model performed the worst among all the models evaluated by the experts, ranking ninth. These results underscore the importance of considering multiple expert opinions and the necessity of a rigorous evaluation process when selecting the optimal model for a robotic hand control system. Table 11 and Fig. 8 present the results of the evaluation benchmarking group using the fusion model, with an external arithmetic mean grouping method applied by the three experts.

The results highlight the importance of evaluating different models for various applications. Among the evaluated models, the RF model outperformed all the other models, as indicated by its performance score (0.21636). On the other hand, the SVM model showed the worst performance, with a performance score of 0.73951. The GB model demonstrated the second-best performance, with a performance score of 0.23735, indicating its potential for high-performance results. However, the LR model had the second-worst performance score, with a value of 0.54907, suggesting the need for improvement in its performance. In conclusion, the benchmarks indicate that the RF model outperforms both standard and feature fusion preprocessing, and the model is considered the top performer among the selected ML models. To this end, the following section focuses on evaluating the RF model's performance using an unseen dataset and clarifying its explainability processes.

4. Methodology evaluation and explainability

In this section, the developed methodology undergoes a comprehensive evaluation across two critical dimensions. First, the selected ML model (RF) is evaluated via new, unseen data to assess its performance and adaptability. The results of this evaluation are presented in detail to highlight the model's robustness and efficiency. Second, the interpretability of the model is explored via the LIME method. Through these evaluations, the aim is to establish the trustworthiness, versatility, and interpretability of the developed methodology. These considerations are dominant in the context of robotic hand control systems based on the BCI technique.

Table 11
Group benchmarking results for the ML models developed on Dataset II.

ML Models	Group Performance Score	Rank
kNN	0.30340	4
DT	0.50802	7
SGD	0.35278	5
SVM	0.73951	9
RF	**0.21636**	**1**
GB	0.23735	2
NN	0.35494	6
AdaBoost	0.30216	3
LR	0.54907	8

4.1. Evaluation of the RF model on the unseen BCI MI dataset (Dataset III)

This section aims to evaluate the chosen ML model (RF), which was selected through the decision-making process applied to an unseen EEG-MI dataset. The experiment focuses solely on adversarial test examples derived from both standard and feature fusion preprocessing methods, assessing the robustness of the RF model. To achieve this, Dataset III was composed of data collected via an EMOTIV EPOC EEG device (specifically, the EPOC + version) at Universiti Pendidikan Sultan Idris (UPSI) by the authors [57].

The recording protocol involved two EEG-MI-based control commands for right and left movements within a simulated environment integrated with an EMOTIV EPOC EEG device. During the recording session, all the subjects were seated in an armchair, facing a flat-screen monitor positioned approximately 1 m away at eye level. The subjects were instructed to keep their hands relaxed and avoid making overt movements. The MI task was initiated by audible cues, each lasting for 8 s, following the Graz protocol.

The dataset comprises EEG data from four subjects (S1, S2, S3, and S4), all of whom had normal vision during the experiments. Each subject performed the task in four sessions, with each session comprising two runs. Each run included 20 trials, resulting in a total of 160 EEG-MI signals recorded for each subject.

The performance of the chosen RF model is assessed via the unseen Dataset III to validate its ability to decode MI signals acquired from a different EEG device. This experiment aims to test the RF model's suitability for addressing the complexity of brain signals and handling the high dimensionality of EEG data from a 14-channel EEG device. The results are presented in Tables 12 and 13, which outline the deployment of the RF model in real-time applications for robotic hand control systems connected to wireless EEG devices, such as the EMOTIV EPOC.

In practice, the results of this experiment are presented in Tables 12 and 13, which demonstrate the ability of the optimal ML model to distinguish between left and right commands. This is evident from the mean Acc, which is 83 % for the RF adversarial model with standard preprocessing and 86 % for the RF adversarial model with feature fusion preprocessing. Notably, the RF model results in a 3 % improvement in the mean Acc for the four subjects when the feature fusion technique is used. Additionally, individual subjects achieve accuracies of 2 %, 4 %, 1 %, and 3 % for S1, S2, S3, and S4, respectively, with feature fusion preprocessing.

In terms of the AUC metric, the mean value for all the subjects reached 89 % with standard preprocessing. It increased to 91 % with feature fusion preprocessing, representing a 2 % improvement achieved through the feature fusion technique. For the F1 metrics, the mean values for all the subjects increased from 84 % with standard preprocessing to 86 % with feature fusion preprocessing, representing a 2 % enhancement achieved through the feature fusion approach. The evaluation metric enhancement with the feature fusion preprocessing technique indicates the model's ability to improve and generalize to unseen data, thereby utilizing new data for the model. The unseen dataset of

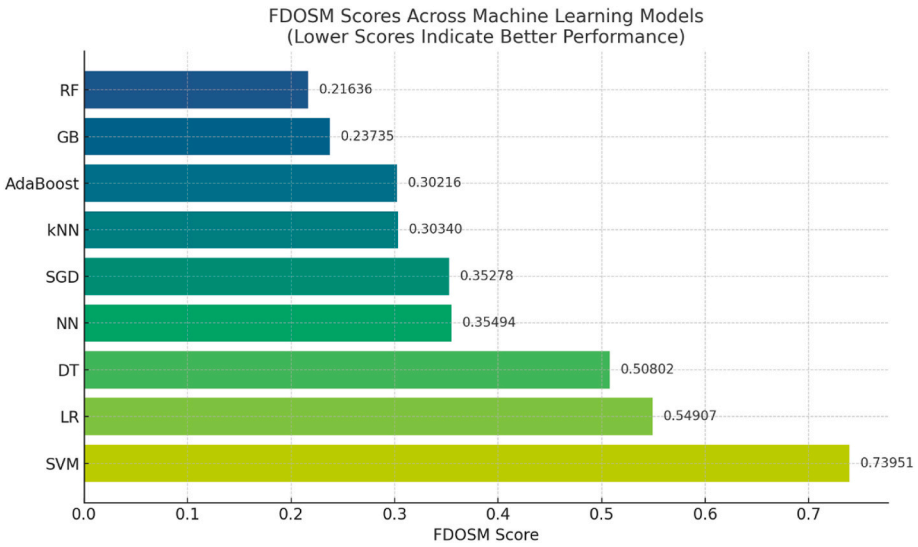


Fig. 8. FDOSM scores across machine learning models (lower scores indicate better performance). This bar chart presents a comparative analysis of the FDOSM scores for various machine learning models applied in the study. Models with lower scores demonstrate superior performance according to the FDOSM evaluation, with RF and GB showing the best results among the tested algorithms.

Table 12
Results of the RF model with adversarial attacks based on standard preprocessing of Dataset III.

Subjects	Test time of unseen data	AUC	Acc	F1	Precision	Recall	LogLoss	Specificity
S1	0.269	0.886	0.801	0.801	0.803	0.801	0.609	0.801
S2	0.347	0.854	0.776	0.786	0.779	0.776	0.610	0.777
S3	0.346	0.904	0.852	0.852	0.852	0.852	0.614	0.852
S4	0.361	0.954	0.911	0.941	0.942	0.941	0.185	0.941
Mean	**0.331**	**0.899**	**0.835**	**0.845**	**0.844**	**0.843**	**0.505**	**0.843**

Table 13
Results of the RF model with adversarial methods on dataset III with feature fusion preprocessing.

Subjects	Test time of unseen data	AUC	CA	F1	Precision	Recall	LogLoss	Specificity
S1	0.184	0.896	0.829	0.829	0.829	0.829	0.586	0.829
S2	0.279	0.868	0.810	0.810	0.810	0.810	0.937	0.820
S3	0.291	0.917	0.866	0.866	0.866	0.866	0.420	0.866
S4	0.288	0.971	0.949	0.949	0.949	0.949	0.221	0.949
Mean	**0.261**	**0.913**	**0.864**	**0.864**	**0.864**	**0.864**	**0.541**	**0.866**

Dataset III was used for testing to ensure the robustness of the model and to obtain more confidence in the model's performance through the use of real data. The RF model's ability lies in its capacity to deploy this model in MI-based control systems, which is consistently accomplished in real-world BCI applications by utilizing unseen datasets for testing, thereby ensuring model robustness beyond experimental situations. This comparative result highlights the model's ability to decode MI brain signals for both the left and right hands, which has practical applications in brain-controlled robotics. Specifically, it can be used as a control command for directing a robotic hand to perform up and down movements. Furthermore, the potential contribution of the RF model to the BCI-MI community lies in its ability to facilitate the deployment of this model in real-time MI-based control systems.

4.2. Explainability of the RF model

To further increase the trustworthiness of the RF model, the evaluation was expanded to include the use of the LIME model for explainability purposes. This section illustrates two scenarios in which the LIME model was employed to explain the performance of the selected RF model, which achieved the highest Acc during training on two different datasets. The first scenario involves adversarial ML on Dataset I with standard preprocessing, in which a white-box approach is applied via the FGSM attack. This attack requires knowledge of the model's architecture and parameters. The idea is to perturb the input data by adding a small amount of noise on the basis of the gradient of the loss with respect to the input. The adversarial example is crafted to maximize the loss, causing misclassification while ensuring that the perturbation remains small enough to be imperceptible. Label-switched adversarial samples were used, which involve assigning incorrect labels to perturbed data samples during training. This type of attack, known as a "poisoning" attack, can weaken models by introducing mislabeled examples and causing the model to misclassify similar inputs during testing. Adversarial examples can be mitigated in a real-world scenario via the adversarial robustness toolbox (ART), which offers several defensive strategies to protect ML models from adversarial attacks, such as adversarial training, defensive distillation, and gradient masking. In this work, adversarial training was employed to train the model with adversarial examples, thereby enhancing model robustness. The second scenario employs adversarial ML on Dataset II with feature fusion preprocessing. These experiments were conducted on a system equipped with an Intel(R) Core i7 processor, 8 GB of RAM, and an NVIDIA GeForce 940MX 12 GB graphics card via the Spider Python editor.

Scenario 1: In this scenario, the LIME model's output explanations

Table 14
LIME output explanation for the RF model on EEG-MI Dataset I.

Multiclass labels of Dataset I	LIME model Intercept	LIME model prediction local	The original RF model prediction Right
Right hand	0.495	0.545	0.70
Left hand	0.504	0.454	0.30
Summation	1	1	1

for the RF model regarding right-hand movement (1) and left-hand movement (2) are presented in Table 14 below:

Table 14 provides data on the intercept, local model prediction, and model prediction for the two class labels in the EEG MI dataset, visualizing right-hand and left-hand movements. This analysis focuses on the higher model prediction, which in this randomly chosen sample relates to right-hand movement. The results show that the intercept of the local LIME model is 0.49546, and its prediction (Prediction_local) is 0.54578, whereas the RF model prediction is 0.70. The sum of the LIME model intercept, LIME model prediction, and local and original RF model predictions must all be 1 to ensure that the probabilities are valid and interpretable. Table 15 presents the LIME coefficient model utilized to obtain the prediction locality for right-hand movement. This prediction is calculated by adding the intercept to the sum of the coefficients, as given in Eq. (18).

$$\text{prediction} = \text{intercept} + \text{sum of the coefficients} \tag{18}$$

By applying Eq. (18), the intercept value is (0.49546), and the total number of coefficients is (−5.80000E-05). Therefore, the result of the LIME prediction for right-hand movement is calculated to be 0.54578.

Table 15 presents the coefficients of the LIME for the dataset features and their ranges, clarifying how each feature affects the prediction of the target class for this local sample. The coefficients present the direction of the relationship between the features and class, which is movement in our case. The coefficients of x7 > 0.34, for example, feature values such as x7 > 0.34, with a coefficient of 0.077066, and 0.23 < x22 ≤ 0.29, with a coefficient of 0.033982, are positively correlated with the movement class, suggesting that these values increase the likelihood of predicting right-hand movement. On the other hand, features such as x28 ≤ 0.42, with a coefficient of −0.00501, exhibit a negative correlation with movement prediction, indicating that they reduce the likelihood of the movement class. Each feature's contribution is thus directional, with some supporting and others opposing the movement prediction. The coefficients provided by LIME reflect a local, instance-specific interpretation of the model's behavior, allowing for an understanding of how the RF model weighs each feature for the particular prediction. This localized explanation is crucial for increasing model transparency, as it reveals the relative importance and influence of features in predicting on a per-sample basis rather than generalizing across the entire dataset.

Table 15
Coefficients of the LIME model for right-hand movement imagination based on Dataset I.

#	Coefficients range	Coefficient's value
1	x7 > 0.34	0.07706
2	x9 > 0.35	−0.0615
3	0.23 < x22≤0.29	0.03398
4	0.30 < x24≤0.37	−0.0325
5	x18≤0.83	−0.0207
6	x6 > 0.32	−0.0204
7	x4 > 0.32	0.01638
8	0.68 < x1≤0.77	0.01273
9	0.15 < x21≤0.19	−0.01081
10	x12 > 0.54	−0.00783
11	x15 > 0.56	−0.00526
12	x5 > 0.19	0.00503
13	x28≤0.42	−0.00501
14	x23 > 0.40	−0.00492
15	0.45 < x13≤0.50	−0.00467
16	8.00 < time duration≤12.00	−0.00455
17	x11 > 0.57	−0.00387
18	x14 > 0.56	−0.00382
19	x8 > 0.27	0.003571
20	0.64 < x3≤0.72	−0.00331
21	x17≤0.67	−0.00301
22	x29≤0.45	0.00288
23	0.41 < x27≤0.45	−0.00267
24	0.87 < x16≤0.90	−0.00263
25	0.14 < x19≤0.18	−0.00242
26	0.36 < x25≤0.41	−0.00122
27	x26≤0.43	−0.00039
28	x20 > 0.26	−0.00038
29	0.38 < x30≤0.42	−9.9E-05
30	x2≤0.83	0.000073
31	x10≤0.36	0.050314
Sum of coefficients		**−5.80000E-05**

The LIME outputs were used to increase trust in the model by breaking down how each feature contributed to predicting right-hand movement for this specific sample. LIME's localized explanations enabled the authors to understand the influence of individual features in a transparent and interpretable manner, revealing which factors the model emphasized in its prediction process. By assigning a coefficient value to each feature on the basis of its contribution to the prediction (as given in Table 15), the LIME output helps clarify which aspects of the input data are most relevant for the prediction of movement. This detailed view of feature impacts reinforced the authors' confidence in the model's logic and assured that the model focused on appropriate variables aligned with domain knowledge. Such explainability helps identify whether the model relies on meaningful neurophysiological signals or potential noise and artifacts. Fig. 9 shows a graphical representation of these explanatory variables to elucidate their respective contributions. Within the plot, the light blue bars situated on the right-hand side signify supportive features that demonstrate a positive correlation with the target variable, which, in this case, is the movement class. Conversely, the blue bars located on the left side indicate features that oppose this positive correlation.

The final explanation displays dataset features along with their values for the current random row sample, as given in Table 16.

Table 16 provides the LIME model feature values of the RF model for hand movement classification on the basis of the standard preprocessing dataset I. Notably, the colors in Table 16 correspond to the same colors used in Fig. 9, which represent the results of predicting right-hand movement via the RF model without feature fusion.

Scenario 2: In this scenario, the output explanation of the LIME results for the RF model with feature fusion preprocessing on Dataset II is presented for right-hand movement (1) and left-hand movement (2), as given in Table 17 below:

Table 17 presents the intercept, local model prediction, and model prediction for the two classes (right- and left-hand movement imagination) within EEG-MI dataset II, which utilizes manifold feature fusion preprocessing. Our focus here is on explaining the results for the higher model prediction, specifically for left-hand movement in this randomly selected sample. The output reveals that the local-level LIME model intercept is 0.47339, and the LIME model prediction stands at 0.55960 (Prediction_local). In comparison, the original RF model prediction equals 0.77.

This model achieves the highest rank when the FDOSM method is used. In this sample, the probability of right-hand movement is represented by the blue bar, which equals 0.23. In contrast, the probability of left-hand movement is denoted by the orange bar, which equals 0.77, resulting in a total prediction probability of 1. Table 18 reports the LIME coefficient model used to obtain the local prediction (0.55960) by adding the intercept to the sum of the coefficients. The intercept value is 0.47339, and the total of the coefficients amounts to 0.08620. Consequently, the LIME prediction is calculated as 0.47339 + 0.08620, resulting in a value of 0.559602.

In Table 18, the coefficients associated with features such as (t-SNE-3

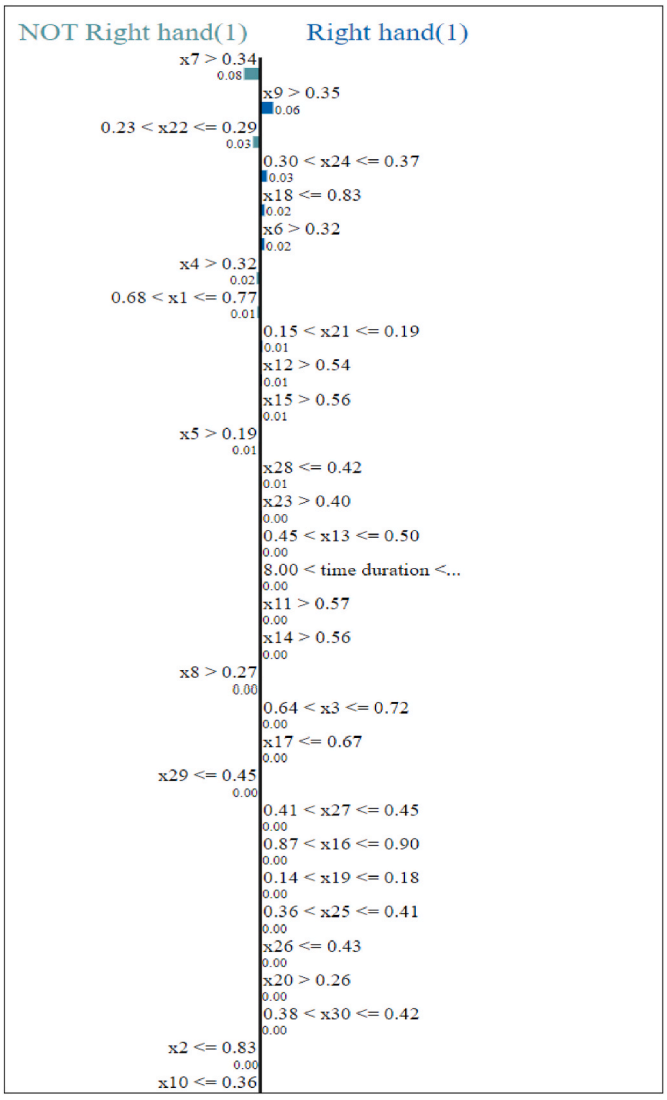


Fig. 9. Illustration of the LIME results for predicting right-hand movement with the RF model without feature fusion.

≤ -4.23) indicate a negative correlation between the values of t-SNE-3 and the hand movement imagination class. Conversely, the coefficients associated with ($-4.79 <$ t-SNE -2≤ 0.04) indicate a positive correlation between the values of t-SNE -2 and the movement class. This analytical approach was systematically applied to comprehend the implications of the remaining features.

Fig. 10 shows a visual representation of the explanatory variables, highlighting their respective contributions. Within the plot, the right-side blue bar represents features that strongly support a positive correlation with the target class, denoting movement level (1). Conversely, the left-hand light blue bars contradict this support, indicating variables that are less influential in predicting the target class.

Table 19 presents the dataset features along with their corresponding values for the current random row sample, providing a comprehensive explanation of the model's prediction.

Table 19 shows the LIME model feature values of the RF model for hand movement imagination via manifold feature fusion. Additionally, the colors in Table 19 correspond to the same colors as those in Fig. 10, which represent the prediction of left-hand movement imagery via the RF model with manifold feature fusion. Both Scenario 1 and Scenario 2 employ LIME as a tool to provide insights into the predictions of the RF model. These scenarios diverge in terms of datasets, preprocessing techniques, model predictions, and the resulting LIME model explanations. Scenario 1 involves predicting right- and left-hand movement imagination with an RF model via standard preprocessing. The model supports the prediction of right-hand movement (70 %). In contrast, in Scenario 2, a different dataset with feature fusion preprocessing was used, resulting in a stronger RF model prediction for left-hand movement (77 %). LIME works to interpret this prediction and investigate the features associated with the coefficients and directions that drive the model's decision.

Table 16
LIME model feature values for the current running sample.

Dataset I Features	Value
x7	0.36
x9	0.38
x22	0.25
x24	0.31
x18	0.8
x6	0.58
x4	0.4
x1	0.72
x21	0.17
x12	0.55
x15	0.69
x5	0.34
x28	0.42
x23	0.49
x13	0.47
time duration	11
x11	0.67
x14	0.69
x8	0.38
x3	0.69
x17	0.27
x29	0.32
x27	0.42
x16	0.88
x19	0.15
x25	0.4
x26	0.32
x20	0.34
x30	0.39
x2	0.79

Table 17
LIME output explanation of the RF model on EEG-MI Dataset II.

Multiclass labels of Dataset II	LIME model Intercept	LIME model prediction local	The original random forest model prediction Right
Right hand	0.526	0.440	0.23
Left hand	0.473	0.559	0.77
Summation	1	1	1

Table 18
Coefficients of the LIME model for right-hand movement on Dataset II.

#	Coefficients range	Coefficient's value
1	t-SNE -3 ≤ -4.23	0.03479
2	$-4.79 <$ t-SNE -2≤ 0.04	0.01717
3	t-SNE -5 ≤ -0.29	0.01590
4	$-0.21 <$ t-SNE -9 ≤ -0.04	0.01371
5	$-0.04 <$ t-SNE -6≤ 0.80	0.01034
6	$-0.37 <$ t-SNE -4≤ 1.34	−0.0085
7	$-7.74 <$ t-SNE -1 ≤ -0.18	−0.0068
8	$-0.21 <$ t-SNE -7 ≤ -0.00	0.00551
9	t-SNE -8 ≤ -0.09	0.00414
Sum of coefficients		**0.086204**

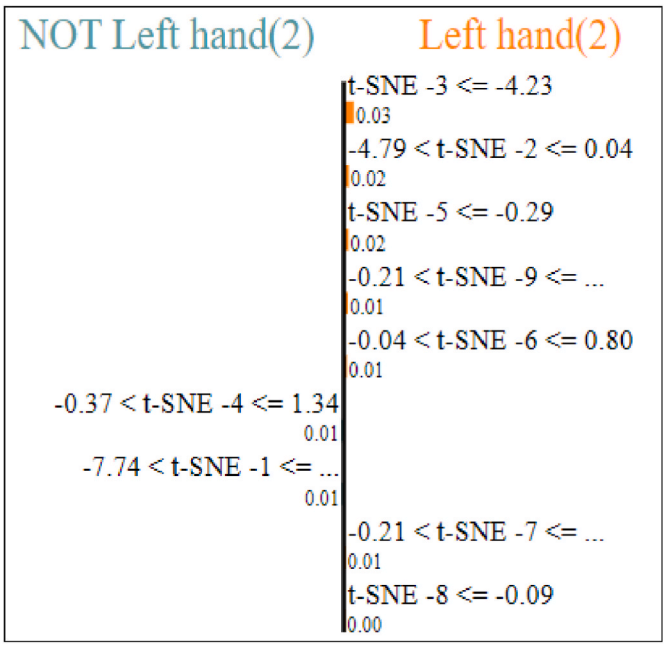


Fig. 10. Visualization of the LIME results for predicting left-hand movement via the RF model with manifold feature fusion.

Table 19
LIME model feature values for the current running sample.

Dataset II Features	Value
t-SNE -3	−13.13
t-SNE -2	−4.2
t-SNE -5	−0.7
t-SNE -9	−0.15
t-SNE -6	0.16
t-SNE -4	1.24
t-SNE -1	−2.72
t-SNE -7	−0.05
t-SNE -8	−0.21

5. Conclusion

To ensure the trustworthiness and security of ML models against adversarial attacks in MI-based robotic hand control systems, extensive efforts have been devoted to enhancing the reliability of multiple ML models for this significant disability-based application, considering both standard and feature fusion preprocessing techniques. The results obtained from these models clearly demonstrate the substantial performance improvement achieved through feature fusion, specifically when t-SNE is employed. This improvement highlights the crucial role of fusion methodologies in designing and evaluating models for complex and demanding environments, such as robotic hand control. Furthermore, the fusion approach enables a comprehensive evaluation that accounts for various criteria and perspectives, ultimately leading to a more resilient and dependable decision-making process. Nonetheless, it is crucial to acknowledge that the fusion approach introduces potential biases and uncertainties, which necessitate careful consideration and effective mitigation strategies.

Furthermore, the FDOSM approach has been leveraged to facilitate the selection of the most suitable ML models, considering both standard test scenarios and adversarial attacks. Despite the wide array of criteria and alternatives, the integration of MPDM and FDOSM has proven to be a successful strategy for evaluating and benchmarking the nine ML models with respect to the nine evaluation metrics, serving as criteria for the two adversarial perspectives. This study employs three datasets in its experimental methodology. The most promising results emerge when the ML models developed on Dataset III are considered, particularly in the context of feature fusion and when adversarial test examples are applied. These findings confirm the validity of the study's hypothesis that feature fusion preprocessing can significantly enhance the performance of ML models for robotic hand control systems.

This study advocates that stakeholders and developers adopt the developed methodology as a prerequisite before producing their robotic hand control systems. This underscores the importance of selecting the optimal ML models, which should not rely solely on standard preprocessing approaches and normal adversarial scenarios; instead, a clear and well-defined decision-making methodology is imperative.

Finally, owing to the inherent characteristics of the EEG signal itself, it is challenging to achieve ideal classification accuracy for right- and left-hand imagery via the nonfusion approach. However, using the integrated evaluation framework with the fusion approach helps in effectively recognizing the MI brain signal with better classification accuracy. Additionally, a trustworthy BCI-based disability application utilizing MI brain signals can be deployed in the BCI community in accordance with the developed robust evaluation framework. Further investigation is needed to address the challenges and complexities associated with developing trustworthy ML models using multiclass MI signals or other BCI modalities, such as P300 and SSVEP, via the fusion technique. Multiple performance samples, typically obtained through repeated cross-validation, are required for statistical tests (such as ANOVA and t tests). Significance testing was not applicable because each model was only run once on a fixed train–test split. To confirm whether metric differences are actually significant, future research will include repeated cross-validation and related significance tests.

CRediT authorship contribution statement

A.S. Albahri: Writing – original draft, Visualization, Validation, Data curation. **Rula A. Hamid:** Investigation, Funding acquisition, Formal analysis. **M.E. Alqaysi:** Writing – original draft, Visualization, Validation, Supervision. **Z.T. Al-Qaysi:** Supervision, Software, Resources. **O.S. Albahri:** Supervision, Project administration, Methodology. **A.H. Alamoodi:** Writing – original draft, Supervision, Data curation, Conceptualization. **Raad Z. Homod:** Supervision, Project administration, Methodology. **Muhammet Deveci:** Resources, Project administration. **Iman Mohamad Sharaf:** Methodology, Investigation, Formal analysis.

Ethics statement

This research followed ethical guidelines for research involving human subjects, the first dataset used in this study was publicly accessible and did not require informed consent. For the second dataset, all participants provided informed consent prior to their participation. Data collected was anonymized to ensure confidentiality.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2025.110922.

Data availability

The first dataset generated and/or analyzed during the current study is available at https://www.bbci.de/competition/iv/desc_2b.pdf. The

second dataset is available from the corresponding author upon reasonable request.

References

[1] Q.K. Pham, T.V. Vo, P.T. Tran, On the implementation of a low-cost mind-voice-and-gesture-controlled humanoid robotic arm using leap motion and Neurosky sensor, J. Electr. Eng. Technol. 17 (1) (2022) 665–683, https://doi.org/10.1007/s42835-021-00903-5.

[2] H.M.K.K.M.B. Herath, W.R. De Mel, Controlling an anatomical robot hand using the brain-computer interface based on motor imagery, Adv. Human-Computer Interact. 2021 (2021) 1–15, https://doi.org/10.1155/2021/5515759.

[3] J.M. Górriz, et al., Computational approaches to explainable artificial intelligence: advances in theory, applications and trends, Inf. Fusion 100 (2023) 101945, https://doi.org/10.1016/j.inffus.2023.101945.

[4] S.U. Amin, M. Alsulaiman, G. Muhammad, M.A. Bencherif, M.S. Hossain, Multilevel weighted feature fusion using convolutional neural networks for EEG motor imagery classification, IEEE Access 7 (2019) 18940–18950, https://doi.org/10.1109/ACCESS.2019.2895688.

[5] A. Sarıkaya, B.G. Kılıç, M. Demirci, RAIDS: robust autoencoder-based intrusion detection system model against adversarial attacks, Comput. Secur. 135 (Sep. 2023) 103483, https://doi.org/10.1016/j.cose.2023.103483.

[6] M. Kianpour, S.-F. Wen, Timing attacks on machine learning: state of the art, Intell. Sys. App.: Proc. 2019 Intell. Sys. Conf. (IntelliSys) 1 (2020) 111–125.

[7] H. Xu, et al., Adversarial attacks and defenses in images, graphs and text: a review, Int. J. Autom. Comput. 17 (2) (2020) 151–178, https://doi.org/10.1007/s11633-019-1211-x.

[8] S.K. Jagatheesaperumal, M. Rahouti, K. Ahmad, A. Al-Fuqaha, M. Guizani, The duo of artificial intelligence and big data for Industry 4.0: applications, techniques, challenges, and future research directions, IEEE Internet Things J. 9 (15) (2022) 12861–12885, https://doi.org/10.1109/JIOT.2021.3139827.

[9] K. Yan, L. Yang, Z. Yang, W. Ren, Enhancing adversarial robustness through stable adversarial training, Symmetry (Basel). 16 (10) (2024) 1363.

[10] P.-Y. Chen, S. Liu, Holistic adversarial robustness of deep learning models, Proc. AAAI Conf. Artif. Intell. 37 (13) (2023) 15411–15420.

[11] P. Kang, J. Li, S. Jiang, P.B. Shull, Reduce system redundancy and optimize sensor disposition for EMG-IMU multimodal fusion human–machine interfaces with XAI, IEEE Trans. Instrum. Meas. 72 (2023) 1–9, https://doi.org/10.1109/TIM.2022.3232159.

[12] H. Qiu, T. Dong, T. Zhang, J. Lu, G. Memmi, M. Qiu, Adversarial attacks against network intrusion detection in IoT systems, IEEE Internet Things J. 8 (13) (2021) 10327–10335, https://doi.org/10.1109/JIOT.2020.3048038.

[13] A. Rahman, et al., Multimodal EEG and Keystroke Dynamics based biometric system using machine learning algorithms, IEEE Access 9 (2021) 94625–94643, https://doi.org/10.1109/ACCESS.2021.3092840.

[14] A.H. Sodhro, C. Sennersten, A. Ahmad, Toward cognitive Authentication for smart healthcare applications, Sensors 22 (6) (2022) 2101, https://doi.org/10.3390/s22062101.

[15] K. Ahmad, M. Maabreh, M. Ghaly, K. Khan, J. Qadir, A. Al-Fuqaha, Developing future human-centered smart cities: critical analysis of smart city security, Data management, and Ethical challenges, Comput. Sci. Rev. 43 (2022) 100452, https://doi.org/10.1016/j.cosrev.2021.100452.

[16] D. Collaris, J.J. Van Wijk, StrategyAtlas: strategy analysis for machine learning interpretability, IEEE Trans. Vis. Comput. Graph. 29 (6) (2023) 2996–3008, https://doi.org/10.1109/TVCG.2022.3146806.

[17] F. Ullah, J. Moon, H. Naeem, S. Jabbar, Explainable artificial intelligence approach in combating real-time surveillance of COVID19 pandemic from CT scan and X-ray images using ensemble model, J. Supercomput. 78 (17) (2022) 19246–19271, https://doi.org/10.1007/s11227-022-04631-z.

[18] M.E. Alqaysi, A.S. Albahri, R.A. Hamid, Diagnosis-based hybridization of multimedical tests and sociodemographic characteristics of autism spectrum disorder using artificial intelligence and machine learning techniques: a systematic review, Int. J. Telemed. Appl. 2022 (2022), https://doi.org/10.1155/2022/3551528.

[19] T. Triwiyanto, et al., Embedded machine learning using a multi-thread algorithm on a Raspberry pi platform to improve prosthetic hand performance, Micromachines 13 (2) (2022) 191, https://doi.org/10.3390/mi13020191.

[20] C. Zeng, S. Li, Z. Chen, C. Yang, F. Sun, J. Zhang, Multifingered robot hand compliant manipulation based on vision-based demonstration and adaptive force control, IEEE Trans. Neural Networks Learn. Syst. 34 (9) (2023) 5452–5463, https://doi.org/10.1109/TNNLS.2022.3184258.

[21] S.R. Nayak, D.R. Nayak, U. Sinha, V. Arora, R.B. Pachori, Application of deep learning techniques for detection of COVID-19 cases using chest X-ray images: a comprehensive study, Biomed. Signal Process Control 64 (2021) 102365, https://doi.org/10.1016/j.bspc.2020.102365.

[22] W. He, et al., SoK: context sensing for access control in the adversarial home IoT, in: Proceedings - 2021 IEEE European Symposium on Security and Privacy, Euro S and P 2021, 2021, pp. 37–53, https://doi.org/10.1109/EuroSP51992.2021.00014.

[23] S. Mittal, S. Tyagi, Computational techniques for real-time credit card fraud detection, Handb. Comput. Networks Cyber Secur. Princ. Paradig. (2019) 653–681, https://doi.org/10.1007/978-3-030-22277-2_26.

[24] M.E. Alqaysi, A.S. Albahri, R.A. Hamid, Hybrid diagnosis models for autism patients based on medical and sociodemographic features using machine learning and multicriteria decision-making (MCDM) techniques: an evaluation and benchmarking framework, Comput. Math. Methods Med. 2022 (2022) 9410222, https://doi.org/10.1155/2022/9410222.

[25] A.H. Alamoodi, et al., Systematic review of MCDM approach applied to the medical case studies of COVID-19: trends, bibliographic analysis, challenges, motivations, recommendations, and future directions, Complex Intell. Syst. 9 (4) (2023) 4705–4731, https://doi.org/10.1007/s40747-023-00972-1.

[26] M. Piwowarski, U.S. Singh, K. Nermend, Application of EEG metrics in the decision-making process, in: Springer Proceedings in Business and Economics, 2020, pp. 187–199, https://doi.org/10.1007/978-3-030-30251-1_14.

[27] M.E. Alqaysi, A.S. Albahri, R.A. Hamid, Evaluation and benchmarking of hybrid machine learning models for autism spectrum disorder diagnosis using a 2-tuple linguistic neutrosophic fuzzy sets-based decision-making model, Neural Comput. Appl. 36 (29) (2024) 18161–18200, https://doi.org/10.1007/s00521-024-09905-6.

[28] A.H. Alamoodi, et al., Based on neutrosophic fuzzy environment: a new development of FWZIC and FDOSM for benchmarking smart e-tourism applications, Complex Intell. Syst. 8 (4) (Aug. 2022) 3479–3503, https://doi.org/10.1007/s40747-022-00689-7.

[29] M.A. Ahmed, et al., Intelligent decision-making framework for evaluating and benchmarking hybridized multi-deep transfer learning models: managing COVID-19 and beyond, Int. J. Inf. Technol. Decis. Mak. (Apr. 2023), https://doi.org/10.1142/S0219622023500463.

[30] Z.T. Al-Qaysi, A.S. Albahri, M.A. Ahmed, S.M. Mohammed, Development of hybrid feature learner model integrating FDOSM for golden subject identification in motor imagery, Phys. Eng. Sci. Med. 46 (4) (2023) 1519–1534, https://doi.org/10.1007/s13246-023-01316-6.

[31] M.S. Al-Samarraay, et al., Extension of interval-valued Pythagorean FDOSM for evaluating and benchmarking real-time SLRSs based on multidimensional criteria of hand gesture recognition and sensor glove perspectives[Formula presented], Appl. Soft Comput. 116 (Feb. 2022) 108284, https://doi.org/10.1016/j.asoc.2021.108284.

[32] A.S. Albahri, et al., Explainable artificial intelligence multimodal of autism triage levels using fuzzy approach-based multicriteria decision-making and LIME, Int. J. Fuzzy Syst. 26 (1) (Nov. 2024) 274–303, https://doi.org/10.1007/s40815-023-01597-9.

[33] Worood Esam Noori, A.S. Albahri, Toward trustworthy myopia detection: integration methodology of deep learning approach, XAI visualization, and user interface system, Appl. Data Sci. Anal. 2023 (Feb. 2023) 1–15, https://doi.org/10.58496/adsa/2023/001. SE-Articles.

[34] A. Barredo Arrieta, et al., Explainable Artificial Intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI, Inf. Fusion 58 (Jun. 2020) 82–115, https://doi.org/10.1016/j.inffus.2019.12.012.

[35] A.S. Albahri, et al., A systematic review of trustworthy and explainable artificial intelligence in healthcare: assessment of quality, bias risk, and data fusion, Inf. Fusion 96 (Aug. 2023) 156–191, https://doi.org/10.1016/j.inffus.2023.03.008.

[36] M.T. Ribeiro, S. Singh, C. Guestrin, Why should I trust you?' Explaining the predictions of any classifier, in: NAACL-HLT 2016 - 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Proceedings of the Demonstrations Session, 2016, pp. 97–101, https://doi.org/10.18653/v1/n16-3020.

[37] Q. Huang, M. Yamada, Y. Tian, D. Singh, Y. Chang, GraphLIME: local interpretable model explanations for graph neural networks, IEEE Trans. Knowl. Data Eng. 35 (7) (2023) 6968–6972, https://doi.org/10.1109/TKDE.2022.3187455.

[38] M. Tangermann, et al., Review of the BCI competition IV, Front. Neurosci. (JULY) (2012) 55, https://doi.org/10.3389/fnins.2012.00055.

[39] I.I. Turulin, H.S. Mogheer, Method and algorithm for synthesis of controlled digital low-pass Butterworth filters on the example of A 4thOrder filter, Proc. Int. Ural Conf. Measure. UralCon 2022-Septe (2022) 336–340, https://doi.org/10.1109/UralCon54942.2022.9906684.

[40] S. Shao, G. Han, T. Wang, C. Lin, C. Song, C. Yao, EEG-based mental workload classification method based on hybrid deep learning model under IoT, IEEE J. Biomed. Heal. Inform. 28 (5) (2024) 2536–2546, https://doi.org/10.1109/JBHI.2023.3281793.

[41] Z. Liu, L. Wang, S. Xu, K. Lu, A multiwavelet-based sparse time-varying autoregressive modeling for motor imagery EEG classification, Comput. Biol. Med. 155 (2023) 106196, https://doi.org/10.1016/j.compbiomed.2022.106196.

[42] D. Avci, E. Sert, E. Dogantekin, O. Yildirim, R. Tadeusiewicz, P. Plawiak, A new super resolution Faster R-CNN model based detection and classification of urine sediments, Biocybern. Biomed. Eng. 43 (1) (2023) 58–68, https://doi.org/10.1016/j.bbe.2022.12.001.

[43] B. Gosala, P. Dindayal Kapgate, P. Jain, R. Nath Chaurasia, M. Gupta, Wavelet transforms for feature engineering in EEG data processing: an application on Schizophrenia, Biomed. Signal Process Control 85 (2023) 104811, https://doi.org/10.1016/j.bspc.2023.104811.

[44] L.P.A. Arts, E.L. van den Broek, The fast continuous wavelet transformation (fCWT) for real-time, high-quality, noise-resistant time–frequency analysis, Nat. Comput. Sci. 2 (1) (2022) 47–58, https://doi.org/10.1038/s43588-021-00183-z.

[45] T. Wu, X. Kong, Y. Zhong, L. Chen, Automatic detection of abnormal EEG signals using multiscale features with ensemble learning, Front. Hum. Neurosci. 16 (2022) 943258, https://doi.org/10.3389/fnhum.2022.943258.

[46] D. Zhang, H. Li, J. Xie, D. Li, MI-DAGSC: a domain adaptation approach incorporating comprehensive information from MI-EEG signals, Neural Netw. 167 (2023) 183–198, https://doi.org/10.1016/j.neunet.2023.08.008.

[47] N. Perera, et al., High performance dataframes from parallel processing patterns, Lect. Notes Comput. Sci. 13826 (2023) 291–304, https://doi.org/10.1007/978-3-031-30442-2_22. LNCS.

[48] R. Li, S. Li, C. He, Y. Zhang, X. Jia, L. Zhang, Class-balanced pixel-level self-labeling for domain adaptive semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11593–11603.
[49] K.M.M. Uddin, N. Biswas, S.T. Rikta, S.K. Dey, Machine learning-based diagnosis of breast cancer utilizing feature optimization technique, Comput. Methods Programs Biomed. Updat. 3 (2023) 100098, https://doi.org/10.1016/j.cmpbup.2023.100098.
[50] S. Ali, et al., Explainable artificial intelligence (XAI): what we know and what is left to attain trustworthy artificial intelligence, Inf. Fusion 99 (2023) 101805, https://doi.org/10.1016/j.inffus.2023.101805.
[51] L. Van Der Maaten, Learning a parametric embedding by preserving local structure, in: Artificial Intelligence and Statistics, 2009, pp. 384–391.
[52] M.C. Cieslak, A.M. Castelfranco, V. Roncalli, P.H. Lenz, D.K. Hartline, t-Distributed Stochastic Neighbor Embedding (t-SNE): a tool for eco-physiological transcriptomic analysis, Mar. Genom. 51 (2020) 100723, https://doi.org/10.1016/j.margen.2019.100723.
[53] Y. Wang, H. Huang, C. Rudin, Y. Shaposhnik, Understanding how dimension reduction tools work: an empirical approach to deciphering T-SNE, UMAP, TriMap, and PaCMAP for data visualization, J. Mach. Learn. Res. 22 (1) (2021) 9129–9201.
[54] P. Zhao, et al., On the design of black-box adversarial examples by leveraging gradient-free optimization and operator splitting method, Proc. IEEE Int. Conf. Comp. Vision 2019-Octob (2019) 121–130, https://doi.org/10.1109/ICCV.2019.00021.
[55] Y. Tashiro, Y. Song, S. Ermon, Diversity can be transferred: output diversification for white- and black-box attacks, Adv. Neural Inf. Process. Syst. 2020-Decem (2020) 4536–4548.
[56] Z. Yuan, J. Zhang, Y. Jia, C. Tan, T. Xue, S. Shan, Meta gradient adversarial attack, in: Proceedings of the IEEE International Conference on Computer Vision, 2021, pp. 7728–7737, https://doi.org/10.1109/ICCV48922.2021.00765.
[57] Z.T. Al-Qaysi, et al., A systematic rank of smart training environment applications with motor imagery brain-computer interface, Multimed. Tools Appl. 82 (12) (May 2023) 17905–17927, https://doi.org/10.1007/s11042-022-14118-x.
[58] O. Zughoul, et al., Novel triplex procedure for ranking the ability of software engineering students based on two levels of AHP and group TOPSIS techniques, Int. J. Inf. Technol. Decis. Making 20 (1) (2021) 67–135.
[59] O.S. Albahri, et al., New mHealth hospital selection framework supporting decentralised telemedicine architecture for outpatient cardiovascular disease-based integrated techniques: Haversine-GPS and AHP-VIKOR, J. Ambient Intell. Hum. Comput. 13 (1) (2022) 219–239.
[60] M.S. Al-Samarraay, et al., Extension of interval-valued Pythagorean FDOSM for evaluating and benchmarking real-time SLRSs based on multidimensional criteria of hand gesture recognition and sensor glove perspectives, Appl. Soft Comput. 116 (2022) 108284.